

Yang Hui, Xiangjie Jiuzhang Suanfa, parts 1-3 (clean public components)

English Translation

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Detailed Analysis of the Mathematical Methods in the Nine Chapters

詳解九章算法
(Part I – 上)

BILINGUAL EDITION / 中英雙語版

By Yang Hui 楊輝
Southern Song Dynasty (c. 1238–1298 CE)

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The “Method Source for Square Root Extraction” (開方作法本源圖)

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Contents

1	Introduction / 引言	3
1.1	Historical Context / 歷史背景	3
1.2	Structure of the Work / 全書結構	3
1.3	Editorial Method / 編纂之法	3
1.4	The Method Source for Square Root Extraction / 開方作法本源	3
2	Original Preface / 原序	4
3	Chapter 1: Multiplication and Division / 乘除	5
3.1	Rectangular Field Method / 直田法	5
3.2	Square Field Method / 方田法	5
3.3	Fraction Multiplication / 乘分	5
3.4	Division with Fractions / 除分	6
4	Chapter 2: Field Measurement / 方田	7
4.1	Reduction of Fractions / 約分	7
4.2	Addition of Fractions / 合分	7
4.3	Subtraction and Comparison of Fractions / 課分	7
4.4	Equalizing Fractions / 平分	8
5	Chapter 3: Grain Conversion / 粟米	9
5.1	Direct Exchange / 經率	9
5.2	Grain Conversion Rates / 粟米換算率	9
5.3	Silk and Currency Conversion / 絲錢換算	10
6	Chapter 4: Proportional Distribution / 衰分	11
6.1	The Method of Distribution / 衰分法	11
6.2	The Cow, Horse, and Sheep / 牛馬羊償粟	11
6.3	The Woman Skilled at Weaving / 女子善織	11
7	Chapter 5: Short Width / Root Extraction / 少廣	12
7.1	The Short Width Method / 少廣法	12
8	The Method Source for Square Root Extraction / 開方作法本源圖	13
8.1	The Original Diagram / 原圖	13
8.2	Explanation of the Structure / 結構釋義	13
8.3	Application to Root Extraction / 開方之應用	13
8.4	Jia Xian's Generating Method / 賈憲增乘方求廉法	14
9	Chapter 6: Commercial Calculations / 商功	15
9.1	Volume Conversion Rates / 土方換算率	15
9.2	City Wall and Ditch Volumes / 城垣溝渠	15
9.3	Special Solids / 特殊立體	15
10	Chapter 7: Equitable Distribution / 均輸	17
10.1	The Equitable Distribution Method / 均輸法	17
10.2	The Pursuit Problem / 相遇問題	17
10.3	The Gourd and Melon / 瓜瓠相逢	17
11	Chapter 8: Excess and Deficit / 盈不足	18
11.1	The Excess and Deficit Method / 盈不足法	18
11.2	Excess and Deficit with Exact Amount / 盈朒適足	18
12	Chapter 9: Systems of Linear Equations / 方程	19
12.1	The Fang Cheng Method / 方程法	19
12.2	Cattle and Sheep / 牛羊直金	19
13	Chapter 10: Right Triangles / 勾股	20

13.1	Pythagorean Theorem / 勾股定理	20
13.2	The Vine and the Tree / 葛纏木	20
13.3	The Reed in the Pond / 葭生中央	20
13.4	The Broken Bamboo / 折竹抵地	20
13.5	Circle Inscribed in a Right Triangle / 勾股容圓	21
14	Square and Cube Root Extraction / 開方術	22
14.1	Jia Xian's "Unlocking" Square Root Method / 賈憲釋鎖開平方法	22
14.2	Cube Root Extraction / 開立方術	22
15	Classification of Methods / 纂類	23
15.1	Method Cross-Reference Table / 方法交叉分類表	23
15.2	Yang Hui's Critical Observations / 楊輝批判性論述	23
16	Conclusion / 結論	24

1 Introduction / 引言

《詳解九章算法》乃南宋楊輝所著之重要數學著作。成書於一二六一年，詳解古代《九章算術》之各題算法。

Xiangjie Jiuzhang Suanfa (詳解九章算法, “Detailed Analysis of the Mathematical Methods in the Nine Chapters”) is one of the most important mathematical works of the Southern Song Dynasty, written by the mathematician **Yang Hui** 楊輝 (c. 1238–1298 CE). Completed in 1261, this work provides detailed explanations and commentaries on the classic *Jiuzhang Suanshu* (Nine Chapters on the Mathematical Art).

1.1 Historical Context / 歷史背景

原《九章算術》成書於漢代（約公元前二百年至公元二百年），凡二百四十六題，分九章。至宋代，原文因靖康之變（一一二七年）而散佚，傳抄訛誤，題隱法晦，學者難以入門。

The original *Jiuzhang Suanshu* dates back to the Han Dynasty (c. 200 BCE–200 CE) and contains 246 mathematical problems organized into nine chapters. By the Song Dynasty, the original text had become difficult to understand due to the loss of manuscripts during the *Jingkang Incident* (靖康之變, 1127 CE), corruption of transmitted texts through repeated copying, and obscure problem statements with difficult methods.

輝嘗聞學者，謂九章題問頗隱，法理難明，不得其門而入。於是以答參問，用草考法，因法推類，然後知斯文非古之全經也。

“I have heard scholars say that the problems in the Nine Chapters are rather obscure, and the methods difficult to understand, so one cannot find the door to enter. Therefore, using the answers to examine the questions, employing the worked solutions to investigate the methods, and following the methods to extend by analogy—only then do I know that this text is not the complete classic of antiquity.”

1.2 Structure of the Work / 全書結構

全書凡十二章：一、乘除（四十一題）；二、方田（三十八題）；三、粟米（四十六題）；四、衰分（二十題）；五、少廣（二十四題）；六、商功（二十八題）；七、均輸（二十八題）；八、盈不足（二十題）；九、方程（十八題）；十、勾股（三十八題）；十一、算類。

The original work comprised 12 **chapters**: (1) Multiplication and Division — 41 problems; (2) Field Measurement — 38 problems; (3) Grain Conversion — 46 problems; (4) Proportional Distribution — 20 problems; (5) Short Width / Root Extraction — 24 problems; (6) Commercial Calculations / Engineering — 28 problems; (7) Equitable Distribution — 28 problems; (8) Excess and Deficit — 20 problems; (9) Systems of Linear Equations — 18 problems; (10) Right Triangles — 38 problems; (11) Classification / Compilation.

1.3 Editorial Method / 編纂之法

楊輝解題之法，分為四部：問（設題）、答（給出數值答案）、術（說明一般算法）、草（詳細逐步演算）。此四部結構，為中國數學著述之典範。

Yang Hui's approach to explaining the Nine Chapters was systematic: **Problem (問)** states the mathematical problem; **Answer (答)** gives the numerical answer; **Method (術)** states the general algorithm; **Worked Solution (草)** provides a detailed step-by-step calculation. This four-part structure became a model for Chinese mathematical exposition.

1.4 The Method Source for Square Root Extraction / 開方作法本源

楊輝所錄之開方作法本源圖（今稱楊輝三角或賈憲三角），源出北宋賈憲（約一〇一〇至一〇七〇年）之《釋鎖算書》。其圖列二項式係數至六次冪，用於開方運算。

One of Yang Hui's most celebrated contributions is preserving what we now call **Yang Hui's Triangle** (also known as Pascal's Triangle). In his work, he attributes this to the earlier mathematician **Jia Xian** (c. 1010–1070 CE), who called it the “Method Source for Square Root Extraction” (開方作法本源圖). The triangle provides binomial coefficients used in root extraction. Yang Hui wrote: 出釋鎖算書，賈憲用此術 (“This comes from the *Shisuo Suan Shu*; Jia Xian used this method”).

2 Original Preface / 原序

黃帝九章古序云：國家嘗設算科取士，選九章以為算經之首，蓋猶儒者之六經，醫家之難素，兵法之孫子歟。

昔聖宋紹興戊辰，算士榮榮謂靖康以來，罕有舊本，間有存者，狃於末習，向獲善本，得其全經，復起於學。以魏景元元年劉徽等、唐朝議大夫、行太史令上輕車都尉李淳風等注釋，聖宋右班直賈憲撰草。

輝嘗聞學者，謂九章題問頗隱，法理難明，不得其門而入。於是以答參問，用草考法，因法推類，然後知斯文非古之全經也。將後賢補贅之文，修前代已廢之法，刪立題術，又纂法問，詳著於後，儻得賢者，改而正諸，是所願也。

“The ancient preface to the Yellow Emperor’s Nine Chapters states: The state once established an examination in mathematics for selecting officials, choosing the Nine Chapters as the foremost mathematical classic—this is like the Six Classics for Confucian scholars, the *Nanjing* and *Suwen* for physicians, or the *Art of War* for military strategists.”

“In the past, during the Wuchen year of the Shaoxing reign of our Sacred Song Dynasty, the mathematician Rong Qi said that since the Jingkang period, old copies have been rare, and those occasionally surviving have been corrupted by inferior practices. Having obtained a good copy and recovered the complete classic, study was revived. The annotations were by Liu Hui et al. (Jingyuan first year of Wei) and by Li Chunfeng et al. (Senior Compiler of Tang Dynasty, Acting Grand Astrologer), while the detailed solutions were composed by Jia Xian of the Song Dynasty.”

“I have heard scholars say that the problems in the Nine Chapters are rather obscure, and the principles difficult to understand, so one cannot find the door to enter. Therefore, using the answers to examine the questions, employing the worked solutions to investigate the methods, and following the methods to extend by analogy—only then do I know that this text is not the complete classic of antiquity. Taking the supplementary texts added by later worthies and repairing the methods abandoned by earlier generations, I have reorganized the problem-methods and compiled the method-questions, recording them in detail below. If worthy scholars would correct and improve them, that is my wish.”

3 Chapter 1: Multiplication and Division / 乘除

第一章論乘除之基本運算，凡四十一題，多取方田章及粟米之題。

The first chapter covers fundamental operations of multiplication and division, with 41 problems drawn mostly from the “Field Measurement” (方田) chapter and grain conversion problems.

3.1 Rectangular Field Method / 直田法

術：以廣乘從得積。以畝法（一畝二百四十平方步）除之。

Method: Multiply the width (廣) by the length (從) to obtain the area (積). Divide by the unit conversion (畝法, 1 mu = 240 square bu).

PROBLEM 1

廣十五步，從十六步，問田幾何？答曰：一畝。 A field has width 15 bu and length 16 bu. What is the area? **Answer:** 1 mu.

SOLUTION (草曰):

廣十五步，從十六步。積 = $15 \times 16 = 240$ 平方步。一畝 = 二百四十平方步，故為一畝。

Width = 15 bu, Length = 16 bu. Area = $15 \times 16 = 240$ square bu. Since 1 mu = 240 square bu, the area is **1 mu**.

PROBLEM 2

廣十二步，從十四步，問田幾何？答曰：一百六十八步。 A field has width 12 bu and length 14 bu. What is the area in square bu? **Answer:** 168 square bu.

SOLUTION (草曰):

積 = $12 \times 14 = 168$ 平方步。

Area = $12 \times 14 = 168$ square bu.

3.2 Square Field Method / 方田法

術：方田之積，以里自乘，復以三百七十五畝乘之。

Method: Square the side length to obtain the area in square li. Multiply by 375 mu per square li for conversion.

PROBLEM 3

方田一里，問為田幾何？答曰：三頃七十五畝。 A square field has side 1 li. What is its area? **Answer:** 3 qing 75 mu.

SOLUTION (草曰):

積 = $1 \times 1 = 1$ 平方里。 $1 \times 375 = 375$ 畝 = 三頃七十五畝。

Area = $1 \times 1 = 1$ square li. $1 \times 375 = 375$ mu = **3 qing 75 mu**.

3.3 Fraction Multiplication / 乘分

術：分母乘全步，并子以為實。分母相乘為法。以法除實。

Method: For fractions with whole numbers: multiply each whole number by its denominator and add the numerator. Multiply these results to form the dividend (實). Multiply the denominators to form the divisor (法). Divide the dividend by the divisor.

PROBLEM Fraction Multiplication

田廣七步、四分步之三，從十五步、九分步之五，問：為田幾何？答曰：一百二十步，九分步之五。 A field has width $7\frac{3}{4}$ bu and length $15\frac{5}{9}$ bu. What is the area? **Answer:** $120\frac{5}{9}$ square bu.

SOLUTION (草曰):

廣： $7\frac{3}{4} = \frac{31}{4}$ 步。從： $15\frac{5}{9} = \frac{140}{9}$ 步。
積 = $\frac{31}{4} \times \frac{140}{9} = \frac{4340}{36} = 120\frac{5}{9}$ 平方步。

Width: $7\frac{3}{4} = \frac{31}{4}$ bu. Length: $15\frac{5}{9} = \frac{140}{9}$ bu.
Area = $\frac{31}{4} \times \frac{140}{9} = \frac{4340}{36} = 120\frac{5}{9}$ square bu.

3.4 Division with Fractions / 除分

術：以人數為法，錢數為實。有分者通之。實如法而一。

Method: Use the number of people as the divisor (法) and the amount of money as the dividend (實). If there are fractions, convert them. Divide the dividend by the divisor.

PROBLEM Division with Fractions

七人，均八錢三分錢之一，問人得幾何？答曰：人得一錢二十一分錢之四。Seven people share $8\frac{1}{3}$ qian equally. How much does each person receive? **Answer:** $1\frac{4}{21}$ qian each.

SOLUTION (草曰):

總數： $8\frac{1}{3} = \frac{25}{3}$ 錢。
每人得： $\frac{25}{3} \div 7 = \frac{25}{21} = 1\frac{4}{21}$ 錢。

Total: $8\frac{1}{3} = \frac{25}{3}$ qian.
Each receives: $\frac{25}{3} \div 7 = \frac{25}{21} = 1\frac{4}{21}$ qian.

4 Chapter 2: Field Measurement / 方田

方田章專論面積之計算與分數之運算。凡三十八題，涉及各種田形及分數四則運算。

The Field Measurement chapter focuses on area calculations and fraction operations. It originally contained 38 problems covering various field shapes and operations with fractions.

4.1 Reduction of Fractions / 約分

術：可半者半之。不可半者，副置分母子之數，以少減多，更相減損，求其等也。以等數約之。（即輾轉相除之歐幾里得算法。）

Method: If both numerator and denominator are even, halve them. If not, place the numerator and denominator separately and repeatedly subtract the smaller from the larger (Euclidean algorithm) until the greatest common divisor is found. Divide both by this common measure.

PROBLEM Reduction

十八分之十二。問：約之得幾何？答曰：三分之二。 Reduce $\frac{12}{18}$. **Answer:** $\frac{2}{3}$.

SOLUTION (草曰):

十二與十八之等數： $18 - 12 = 6$, $12 - 6 = 6$, $6 - 6 = 0$. 等數 = 6.
 $\frac{12 \div 6}{18 \div 6} =$
GCD of 12 and 18: $18 - 12 = 6$, $12 - 6 = 6$, $6 - 6 = 0$. GCD = 6.
 $\frac{12 \div 6}{18 \div 6} =$

PROBLEM

九十一分之四十九。問：約之得幾何？答曰：十三分之七。 Reduce $\frac{49}{91}$. **Answer:** $\frac{7}{13}$.

SOLUTION (草曰):

輾轉相除： $91 - 49 = 42$, $49 - 42 = 7$, $42 - 7 \times 6 = 0$. 等數 = 7.
 $\frac{49 \div 7}{91 \div 7} =$
Using the Euclidean algorithm: $91 - 49 = 42$, $49 - 42 = 7$, $42 - 7 \times 6 = 0$. GCD = 7.
 $\frac{49 \div 7}{91 \div 7} =$

4.2 Addition of Fractions / 合分

術：母互乘子，并以為實。母相乘為法。實如法而一。母同者直相從之。

Method: Cross-multiply the denominators with the numerators and add to form the dividend. Multiply the denominators to form the divisor. Divide. If the denominators are the same, simply add the numerators.

PROBLEM

二分之一、三分之二、四分之三、五分之四，合之得幾何？答曰：得二，餘六十分之四十三。 Add $\frac{1}{2} + \frac{2}{3} + \frac{3}{4} + \frac{4}{5}$. **Answer:** $2 \frac{43}{60}$.

SOLUTION (草曰):

互乘： $1 \times 3 \times 4 \times 5 + 2 \times 2 \times 4 \times 5 + 3 \times 2 \times 3 \times 5 + 4 \times 2 \times 3 \times 4 = 60 + 80 + 90 + 96 = 326$.
法： $2 \times 3 \times 4 \times 5 = 120$.
 $\frac{326}{120} = 2 \frac{86}{120} = 2 \frac{43}{60}$

Cross-multiply: $1 \times 3 \times 4 \times 5 + 2 \times 2 \times 4 \times 5 + 3 \times 2 \times 3 \times 5 + 4 \times 2 \times 3 \times 4 = 60 + 80 + 90 + 96 = 326$.
Divisor: $2 \times 3 \times 4 \times 5 = 120$.
 $\frac{326}{120} = 2 \frac{86}{120} = 2 \frac{43}{60}$.

4.3 Subtraction and Comparison of Fractions / 課分

術：母互乘子，以少減多，餘為實。母相乘為法。實如法而一。即兩分數之差也。

Method: Cross-multiply numerators, subtract the smaller from the larger. The remainder is the dividend, the product of denominators is the divisor. This gives the difference (also called “the excess”).

PROBLEM

八分之五，比二十五分之十六。問：孰多多幾何？答曰：二十五分之十六，多多二百分之三。 Compare $\frac{5}{8}$ and $\frac{16}{25}$. Which is larger and by how much? **Answer:** $\frac{16}{25}$ is larger by $\frac{3}{200}$.

SOLUTION (草曰):

$$\frac{16}{25} - \frac{5}{8} = \frac{16 \times 8 - 5 \times 25}{25 \times 8} = \frac{128 - 125}{200} = \frac{3}{200}.$$

故 $\frac{16}{25}$ 大 $3\frac{3}{2000}$.

$$\frac{16}{25} - \frac{5}{8} = \frac{16 \times 8 - 5 \times 25}{25 \times 8} = \frac{128 - 125}{200} = \frac{3}{200}.$$

So $\frac{16}{25}$ is larger by

4.4 Equalizing Fractions / 平分

術：母互乘子，各為列實。并諸列實為平實。母相乘為法。以列數乘未并者，各為列實。以平實減列實，餘為減數；以列實減平實，餘為益數。減多益少，使各平於平實。

Method: Cross-multiply numerators. Sum all numerators for the “level dividend.” Multiply all denominators for the divisor. Multiply the number of terms by each uncrossed numerator for the column dividends. Subtract the level dividend from each column dividend to find what must be subtracted; subtract each column dividend from the level dividend to find what must be added. Combine the subtractions to benefit the smaller values. Divide the level dividend by the divisor to find the equalized value.

PROBLEM

二分之一、三分之二、四分之三，減多益少幾何？而平？答曰：減四分之三，求之者四，減三分之二，求之者一，益二分之一，求之者五，各平於三十六分之二十三。 Given $\frac{1}{2}$, $\frac{2}{3}$, $\frac{3}{4}$, redistribute by subtracting from larger and adding to smaller to make them equal. **Answer:** Subtract $\frac{4}{36}$ from $\frac{3}{4}$; subtract $\frac{1}{36}$ from $\frac{2}{3}$; add $\frac{5}{36}$ to $\frac{1}{2}$. All equalized to $\frac{23}{36}$.

SOLUTION (草曰):

平值 = $\frac{23}{36}$ 。
 $\frac{3}{4} = \frac{27}{36}$: 減 $\frac{4}{36}$ 。
 $\frac{2}{3} = \frac{24}{36}$: 減 $\frac{1}{36}$ 。
 $\frac{1}{2} = \frac{18}{36}$: 益 $\frac{5}{36}$ 。
 皆平於 $23\frac{23}{360}$.

Level value = $\frac{23}{36}$.
 $\frac{3}{4} = \frac{27}{36}$: subtract $\frac{4}{36}$.
 $\frac{2}{3} = \frac{24}{36}$: subtract $\frac{1}{36}$.
 $\frac{1}{2} = \frac{18}{36}$: add $\frac{5}{36}$.
 All equalized to $23\frac{23}{360}$.

Original Chinese (原文)

English Translation (英譯)

5 Chapter 3: Grain Conversion / 粟米

粟米章論各種穀物與商品之比例交換。凡四十六題，其中六題為基本乘除，三十一題為比例交換，九題為比例率。

The Grain Conversion chapter deals with proportional exchange between different types of grain and commodities. It originally contained 46 problems with 6 on basic multiplication/division, 31 on proportional exchange, and 9 on proportional rates.

5.1 Direct Exchange / 經率

術：以所買率為法，以所出錢為實。實如法得一。

Method: Use the money amount as the dividend (實) and the quantity purchased as the divisor (法). Divide the dividend by the divisor.

PROBLEM

錢一百六十文，買瓚甓十八枚，問枚價幾何？答曰：一枚八錢九分錢之八。 With 160 qian, buy 18 roof tiles. What is the price per tile? **Answer:** $8\frac{8}{9}$ qian per tile.

SOLUTION (草曰):

$$160 \div 18 = \frac{160}{18} = \frac{80}{9} = 8\frac{8}{9} \text{ 文/枚。}$$

$$160 \div 18 = \frac{160}{18} = \frac{80}{9} = 8\frac{8}{9} \text{ qian per tile.}$$

5.2 Grain Conversion Rates / 粟米換算率

《九章》立穀物換算之標準率：粟率五十；糲米率三十；粳米率二十七；御米率二十四；麥率四十五；菽率四十五。

The Nine Chapters establishes standard conversion rates between different grains: Millet (粟): rate = 50; Coarse rice (糲米): rate = 30; Fine rice (粳米): rate = 27; Polished rice (御米): rate = 24; Wheat (麥): rate = 45; Soybeans (菽): rate = 45.

術：以所有數乘所求率為實，以所有率為法，實如法而一。

Method: “What you seek multiplied by the given amount, divided by what you have.” (以所有數乘所求率為實，以所有率為法，實如法而一。)

PROBLEM

粟一斗。問為糲米幾何？答曰：六升。 Given 1 dou of millet, how much coarse rice (糲米) does it make? **Answer:** 6 sheng.

SOLUTION (草曰):

$$\text{粟率五十，糲米率三十。} \\ 1 \text{ 斗} \times \frac{30}{50} = \frac{3}{5} \text{ 斗} = 6 \text{ 升。}$$

$$\text{Millet rate} = 50, \text{ coarse rice rate} = 30. \\ 1 \text{ dou} \times \frac{30}{50} = \frac{3}{5} \text{ dou} = 6 \text{ sheng.}$$

PROBLEM

粟四斗五升。問為粳米幾何？答曰：二斗一升五分升之三。 Given 4 dou 5 sheng of millet, how much fine rice (粳米) does it make? **Answer:** 2 dou 1 sheng $\frac{3}{5}$.

SOLUTION (草曰):

$$4.5 \times \frac{27}{50} = \frac{9}{2} \times \frac{27}{50} = \frac{243}{100} = 2.43 \text{ 斗} = \text{二斗一升五分升之三。}$$

$$4.5 \times \frac{27}{50} = \frac{9}{2} \times \frac{27}{50} = \frac{243}{100} = 2.43 \text{ dou} = 2 \text{ dou } 1 \text{ sheng } \frac{3}{5}.$$

PROBLEM

糲米十五斗五升五分升之二。問為粟幾何？答曰：二十五斗九升。 Given 15 dou $5\frac{2}{5}$ sheng of coarse rice, how much millet was the original amount? **Answer:** 25 dou 9 sheng.

SOLUTION (草曰):

$$15\frac{2}{5} \text{ 斗} = \frac{77}{5} \text{ 斗。} \\ \frac{77}{5} \times \frac{50}{30} = \frac{777}{30} = 25.9 \text{ 斗} = \text{二十五斗九升。}$$

$$15\frac{2}{5} \text{ 斗} = \frac{777}{50} \text{ dou.} \\ \frac{777}{50} \times \frac{50}{30} = \frac{777}{30} = 25.9 \text{ dou} = 25 \text{ dou } 9 \text{ sheng.}$$

5.3 Silk and Currency Conversion / 絲錢換算

PROBLEM

絲一斤，價三百四十五，今有七兩一十二銖。問錢，答曰：一百六十一錢，三十二分錢之二十三。 Silk costs 345 qian per jin. If one has 7 liang 12 zhu of silk, what is its value in qian? **Answer:** $161 \frac{23}{32}$ qian.

SOLUTION (草曰):

一斤 = 十六兩 = 三百八十四銖。七兩十二銖 = $7 \times 24 + 12 = 180$ 銖。

值 = $345 \times \frac{180}{384} = 345 \times \frac{15}{32} = \frac{5175}{32} = 161 \frac{23}{32}$ 錢。

$1 \text{ jin} = 16 \text{ liang} = 384 \text{ zhu}$. $7 \text{ liang } 12 \text{ zhu} = 7 \times 24 + 12 = 180 \text{ zhu}$.

$\text{Value} = 345 \times \frac{180}{384} = 345 \times \frac{15}{32} = \frac{5175}{32} = 161 \frac{23}{32} \text{ qian}$.

6 Chapter 4: Proportional Distribution / 衰分

衰分章論按給定比例分配數量之問題。凡二十題。

The Proportional Distribution chapter deals with dividing quantities in given proportions. It originally contained 20 problems.

6.1 The Method of Distribution / 衰分法

術：各置列衰，副并為法。以所分乘未并者，各自為實。實如法而一。不滿法者，以法命之。

Method: Set out the distribution rates (衰). Combine them as the divisor. Multiply what is to be distributed by each uncombined rate to form the column dividends. Divide each by the divisor. If there is a remainder, express it as a fraction over the divisor.

PROBLEM Five Ranks Sharing Deer / 五爵分鹿

大夫、不更、簪裹、上造、公士凡五人，以爵次高下均五鹿，問各幾何？答曰：大夫一鹿，三分鹿之二；不更一鹿，三分鹿之一；簪裹一鹿；上造鹿三分之二；公士鹿三分之一。Five nobles—a Dafu, a Bugeng, a Zanmao, a Shangzao, and a Gongshi—are to share 5 deer in proportion to their ranks (5:4:3:2:1). **Answer:** Dafu $1\frac{2}{3}$; Bugeng $1\frac{1}{3}$; Zanmao 1; Shangzao $\frac{2}{3}$; Gongshi $\frac{1}{3}$ deer.

SOLUTION (草曰):

衰：大夫五、不更四、簪裹三、上造二、公士一。
并 = $5 + 4 + 3 + 2 + 1 = 15$ 。
大夫： $5 \times \frac{5}{15} = \frac{25}{15} = 1\frac{2}{3}$ 鹿。
不更： $5 \times \frac{4}{15} = \frac{20}{15} = 1\frac{1}{3}$ 鹿。
簪裹： $5 \times \frac{3}{15} = 1$ 鹿。
上造： $5 \times \frac{2}{15} = \frac{2}{3}$ 鹿。
公士： $5 \times \frac{1}{15} = \frac{1}{3}$ 鹿。

Rates: Dafu 5, Bugeng 4, Zanmao 3, Shangzao 2, Gongshi 1.
Sum = $5 + 4 + 3 + 2 + 1 = 15$.
Dafu: $5 \times \frac{5}{15} = \frac{25}{15} = 1\frac{2}{3}$ deer.
Bugeng: $5 \times \frac{4}{15} = \frac{20}{15} = 1\frac{1}{3}$ deer.
Zanmao: $5 \times \frac{3}{15} = 1$ deer.
Shangzao: $5 \times \frac{2}{15} = \frac{2}{3}$ deer.
Gongshi: $5 \times \frac{1}{15} = \frac{1}{3}$ deer.

6.2 The Cow, Horse, and Sheep / 牛馬羊償粟

PROBLEM

牛、馬、羊食人苗，苗主責之粟五斗，牛食馬之半，馬食羊之半，欲衰償之？答曰：牛二斗八升，七分升之四；馬一斗四升，七分升之二；羊七升，七分升之一。A cow, a horse, and a sheep ate someone's crops. The owner demands 5 dou of millet as compensation. The cow ate twice what the horse ate, and the horse ate twice what the sheep ate. **Answer:** Cow $2\frac{4}{7}$ dou; Horse $1\frac{2}{7}$ dou; Sheep $\frac{1}{7}$ dou.

SOLUTION (草曰):

衰：牛四、馬二、羊一。
并 = $4 + 2 + 1 = 7$ 。
牛： $5 \times \frac{4}{7} = \frac{20}{7} = 2\frac{6}{7}$ 斗 = 二斗八升 $\frac{4}{7}$ 。
馬： $5 \times \frac{2}{7} = \frac{10}{7} = 1\frac{3}{7}$ 斗 = 一斗四升 $\frac{2}{7}$ 。
羊： $5 \times \frac{1}{7} = \frac{5}{7}$ 斗 = 七升 $\frac{1}{7}$ 。

The rates are: Cow 4, Horse 2, Sheep 1.
Sum = $4 + 2 + 1 = 7$.
Cow: $5 \times \frac{4}{7} = \frac{20}{7} = 2\frac{6}{7}$ dou = 2 dou 8 sheng $\frac{4}{7}$.
Horse: $5 \times \frac{2}{7} = \frac{10}{7} = 1\frac{3}{7}$ dou = 1 dou 4 sheng $\frac{2}{7}$.
Sheep: $5 \times \frac{1}{7} = \frac{5}{7}$ dou = 7 sheng $\frac{1}{7}$.

6.3 The Woman Skilled at Weaving / 女子善織

PROBLEM

女子善織，日自倍，五日織五尺。問日織數。答曰：初日一寸，三十一分寸之十九；二日三寸，三十一分寸之七；三日六寸，三十一分寸之十四；四日一尺二寸，三十一分寸之二十八；五日二尺四寸，三十一分寸之二十五。A woman skilled at weaving doubles her output each day. In 5 days she weaves 5 chi. **Answer:** Day 1: $\frac{50}{31}$ cun; Day 2: $\frac{100}{31}$ cun; Day 3: $\frac{200}{31}$ cun; Day 4: $\frac{400}{31}$ cun; Day 5: $\frac{800}{31}$ cun.

SOLUTION (草曰):

設首日織 x ，則總數 = $x + 2x + 4x + 8x + 16x = 31x = 5$ 尺。
 $x = \frac{5}{31}$ 尺 = $\frac{50}{31}$ 寸 = $1\frac{19}{31}$ 寸。
次日： $\frac{100}{31} = 3\frac{7}{31}$ 寸。
三日： $\frac{200}{31} = 6\frac{14}{31}$ 寸。
四日： $\frac{400}{31} = 12\frac{28}{31}$ 寸。
五日： $\frac{800}{31} = 25\frac{25}{31}$ 寸。

Let the first day's output be x . Then total = $x + 2x + 4x + 8x + 16x = 31x = 5$ chi.
 $x = \frac{5}{31}$ chi = $\frac{50}{31}$ cun = $1\frac{19}{31}$ cun.
Day 2: $\frac{100}{31} = 3\frac{7}{31}$ cun.
Day 3: $\frac{200}{31} = 6\frac{14}{31}$ cun.
Day 4: $\frac{400}{31} = 12\frac{28}{31}$ cun.
Day 5: $\frac{800}{31} = 25\frac{25}{31}$ cun.

7 Chapter 5: Short Width / Root Extraction / 少廣

少廣章論已知矩形面積與一邊而求另一邊，涉及分數除法與開方。凡二十四題。

The Short Width chapter deals with finding an unknown dimension of a rectangle given its area and one side, which leads to problems of division with fractions and root extraction. It originally contained 24 problems.

7.1 The Short Width Method / 少廣法

術：置全步及分子子。各以分母乘其子、全步，以少減多，通分內子，令母互乘。并所有以為法。以畝積二百四十步乘之為實。實如法而一，得從。

Method: Set out the whole steps and the fraction denominators and numerators. Separately multiply the denominators by themselves, then multiply the whole steps and numerators. Divide each by its original denominator and combine. Use this as the divisor. Multiply the whole steps by their combined denominators, then multiply by the field area (mu) in square bu for the dividend. Divide the dividend by the divisor.

PROBLEM Short Width 1

田一畝廣一步半，問從？答曰：一百六十步。 A field of 1 mu has width $1\frac{1}{2}$ bu. What is the length? **Answer:** 160 bu.

SOLUTION (草曰):

廣 = $1\frac{1}{2} = \frac{3}{2}$ 步。積 = 一畝 = 二百四十平方步。
從 = $240 \div \frac{3}{2} = 240 \times \frac{2}{3} = 160$ 步。

Width = $1\frac{1}{2} = \frac{3}{2}$ bu. Area = 1 mu = 240 square bu.
Length = $240 \div \frac{3}{2} = 240 \times \frac{2}{3} = 160$ bu.

PROBLEM

田一畝廣一步半、三分步之一。問從？答曰：一百三十步一十一分步之一十。 A field of 1 mu has width $1\frac{1}{2}\frac{1}{3}$ bu. What is the length? **Answer:** $130\frac{10}{11}$ bu.

SOLUTION (草曰):

廣 = $1 + \frac{1}{2} + \frac{1}{3} = \frac{6+3+2}{6} = \frac{11}{6}$ 步。
從 = $240 \div \frac{11}{6} = 240 \times \frac{6}{11} = \frac{1440}{11} = 130\frac{10}{11}$ 步。

Width = $1 + \frac{1}{2} + \frac{1}{3} = \frac{6+3+2}{6} = \frac{11}{6}$ bu.
Length = $240 \div \frac{11}{6} = 240 \times \frac{6}{11} = \frac{1440}{11} = 130\frac{10}{11}$ bu.

PROBLEM

田一畝廣一步半、三分步之一、四分步之一。問從答曰：一百一十五步五分步之一。 A field of 1 mu has width $1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4}$ bu. **Answer:** $115\frac{1}{5}$ bu.

SOLUTION (草曰):

廣 = $1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} = \frac{12+6+4+3}{12} = \frac{25}{12}$ 步。
從 = $240 \div \frac{25}{12} = \frac{2880}{25} = 115\frac{1}{5}$ 步。

Width = $1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} = \frac{12+6+4+3}{12} = \frac{25}{12}$ bu.
Length = $240 \div \frac{25}{12} = \frac{2880}{25} = 115\frac{1}{5}$ bu.

PROBLEM

田一畝廣一步半、三分步之一、四分步之一、五分步之一、六分步之一、七分步之一。問從答曰：九十二步一百二十一分步之六十八。 A field of 1 mu has width $1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \frac{1}{5} + \frac{1}{6} + \frac{1}{7}$ bu. **Answer:** $92\frac{68}{121}$ bu.

SOLUTION (草曰):

用逆少廣術：

廣 = $\frac{420+210+140+105+84+70+60}{420} = \frac{1089}{420}$ 。
約之：等數三，得 $\frac{363}{140}$ 。
從 = $240 \times \frac{140}{363} = \frac{33600}{363} = 92\frac{204}{363} = 92\frac{68}{121}$ 步。

Using the reverse combination-of-fractions method:

Width = $\frac{420+210+140+105+84+70+60}{420} = \frac{1089}{420}$.
Simplifying: $\text{GCD}(1089, 420) = 3$, so $\frac{363}{140}$.
Length = $240 \times \frac{140}{363} = \frac{33600}{363} = 92\frac{204}{363} = 92\frac{68}{121}$ bu.

8 The Method Source for Square Root Extraction / 開方作法本源圖

楊輝書中最著名者，為其所錄之開方作法本源圖（今稱楊輝三角）。楊輝稱此圖源出北宋賈憲（約一〇一〇至一〇七〇年）。

One of the most celebrated parts of Yang Hui's work is his recording of what we now call **Yang Hui's Triangle** (also known as Pascal's Triangle in the West). Yang Hui attributed this to Jia Xian (c. 1010–1070 CE), who called it the “Method Source for Square Root Extraction” (開方作法本源).

8.1 The Original Diagram / 原圖

楊輝釋此圖曰：

左表乃積數，右表乃隅算，中藏者皆廉，以廉乘商方，命實而除之。

Yang Hui presented the triangle with the following explanation: “The left diagonal consists of the accumulated numbers; the right diagonal consists of the corner calculations. Those hidden in the middle are all the ‘flanks’ (intermediate coefficients). Using the flanks, multiply the quotient square, and command the dividend to be divided by it.”

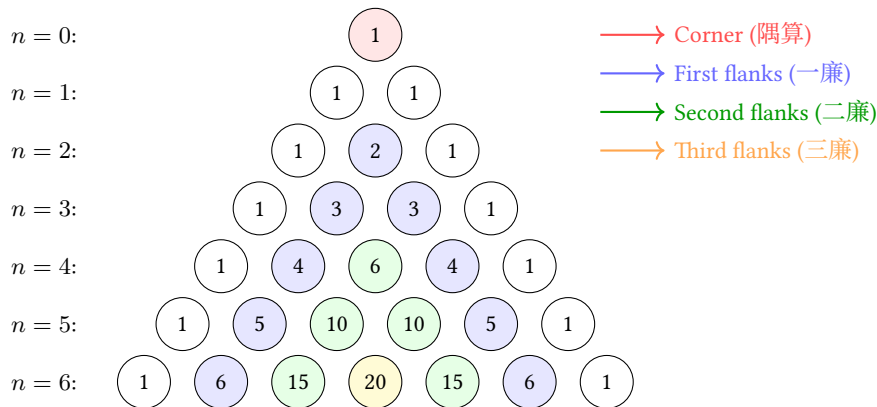


Fig. 1: Yang Hui's Triangle showing binomial coefficients up to the 6th power / 楊輝三角至六次冪

8.2 Explanation of the Structure / 結構釋義

左表（積數）：斜邊之常數項（皆為一），即 $(x + a)^n$ 展開式之常數係數。

右表（隅算）：最高次冪之係數（亦皆為一）。

中藏者（廉）：中間各數為廉，即各中間項之係數。

Left diagonal (左表): The constant terms (1, 1, 1, ...) represent the “accumulated numbers” (積數)—the coefficients of the constant term in $(x + a)^n$.

Right diagonal (右表): The coefficients of the highest power—also all 1's—represent the “corner calculations” (隅算).

Interior numbers (中藏者): The intermediate numbers are the “flanks” (廉)—the coefficients of the intermediate terms.

每行給出 $(x + a)^n$ 展開之係數：

$$(x + a)^0 = 1$$

$$(x + a)^1 = x + a$$

$$(x + a)^2 = x^2 + 2xa + a^2$$

$$(x + a)^3 = x^3 + 3x^2a + 3xa^2 + a^3$$

Each row gives the coefficients for expanding $(x + a)^n$:

$$(x + a)^0 = 1$$

$$(x + a)^1 = x + a$$

$$(x + a)^2 = x^2 + 2xa + a^2$$

$$(x + a)^3 = x^3 + 3x^2a + 3xa^2 + a^3$$

8.3 Application to Root Extraction / 開方之應用

楊輝與賈憲用此係數進行開平方與開立方。開平方之術：一、置積為實。二、商估首位。三、以三角之係數為方法、廉法、隅法。四、乘減遞推。

Yang Hui and Jia Xian used these coefficients to perform square root and cube root extraction. To extract $\sqrt{A}$: (1) Set A as the dividend (實). (2) Estimate the first digit of the root as the quotient (商). (3) Use the coefficients from the triangle to form the “method” (法), “flank” (廉), and “corner” (隅). (4) Multiply and subtract iteratively.

8.4 Jia Xian's Generating Method / 賈憲增乘方求廉法

增乘方求廉法：以一次乘二次，以二次乘三次，遞增一位，至求終。

此法由前行生成後行，相鄰兩數相加得之—即今所謂帕斯卡法則：

$$\binom{n}{k} = \binom{n-1}{k-1} + \binom{n-1}{k}$$

“To seek the flanks by augmented multiplication: multiply once by twice, multiply twice by thrice, increasing by one position each time, until the end is reached.”

This method generates each row from the previous one by adding adjacent entries—exactly the rule we now call Pascal's rule:

$$\binom{n}{k} = \binom{n-1}{k-1} + \binom{n-1}{k}$$

9 Chapter 6: Commercial Calculations / 商功

商功章論工程土方之體積計算。凡二十八題。

The Commercial Calculations chapter deals with volume computations for engineering projects (earthworks, construction, etc.). It originally contained 28 problems.

9.1 Volume Conversion Rates / 土方換算率

《九章》立不同土質之換算率：穿地為基準；壤（鬆土）與穿地之比為 4 : 5；堅（堅土）與穿地之比為 3 : 4。

The Nine Chapters establishes conversion rates between different soil conditions: Excavated earth (穿地): base measure; Loose earth (壤): 4 : 5 ratio with excavated; Compacted earth (堅): 3 : 4 ratio with excavated.

PROBLEM

穿地積一萬尺。問為堅、壤各幾何？答曰：為堅七千五百尺，為壤一萬二千五百尺。An excavation of 10,000 cubic chi. How much compacted and loose earth does it produce? **Answer:** Compacted 7,500; Loose 12,500 cubic chi.

SOLUTION (草曰):

穿地: 壤: 堅 = 4 : 5 : 3。
壤 = $10000 \times \frac{5}{4} = 12500$ 立方尺。
堅 = $10000 \times \frac{3}{4} = 7500$ 立方尺。

Excavated : Loose : Compacted = 4 : 5 : 3.
Loose = $10000 \times \frac{5}{4} = 12,500$ cubic chi.
Compacted = $10000 \times \frac{3}{4} = 7,500$ cubic chi.

9.2 City Wall and Ditch Volumes / 城垣溝渠

術：並上下廣而半之，以高（或深）乘之，又以袤乘之，得積。

Method: For city walls, dikes, ditches, and canals: average the top and bottom widths, multiply by height (or depth), then multiply by length.

PROBLEM

城下廣四丈，上廣二丈，高五丈，袤一百二十六丈五尺。問為尺幾何？答曰：一百八十九萬七千五百尺。A city wall has bottom width 4 zhang, top width 2 zhang, height 5 zhang, and length 126 zhang 5 chi. **Answer:** 1,897,500 cubic chi.

SOLUTION (草曰):

平均廣 = $\frac{4+2}{2} = 3$ 丈。
積 = $3 \times 5 \times 126.5 = 1897.5$ 立方丈 = **1897500** 立方尺。

Average width = $\frac{4+2}{2} = 3$ zhang.
Volume = $3 \times 5 \times 126.5 = 1897.5$ cubic zhang = $1897.5 \times 1000 = 1,897,500$ cubic chi.

9.3 Special Solids / 特殊立體

The Qian Du (塹堵)

術：廣乘袤，又以高乘之，半之。

Method: Multiply length by width, then multiply by height, and divide by 2.

PROBLEM

塹堵下廣二丈，袤十八丈六尺，高二丈五尺，問積尺幾何？答曰：四萬六千五百尺。A *qiandu* has bottom width 2 zhang, length 18 zhang 6 chi, and height 2 zhang 5 chi. **Answer:** 46,500 cubic chi.

SOLUTION (草曰):

積 = $\frac{1}{2} \times 20 \times 186 \times 25 = 46500$ 立方尺。

Volume = $\frac{1}{2} \times 20 \times 186 \times 25 = 46,500$ cubic chi.

The Yang Ma (陽馬)

術：廣乘袤，又以高乘之，三而一。

Method: Multiply length by width, then multiply by height, and divide by 3.

PROBLEM

陽馬廣五尺，袤七尺，高八尺，問積尺幾何？答曰：九十三尺，三分尺之一。A *yangma* has width 5 chi, length 7 chi, and height 8 chi. **Answer:** $93\frac{1}{3}$ cubic chi.

SOLUTION (草曰):

積 = $\frac{1}{3} \times 5 \times 7 \times 8 = \frac{280}{3} = 93\frac{1}{3}$ 立方尺。

Volume = $\frac{1}{3} \times 5 \times 7 \times 8 = \frac{280}{3} = 93\frac{1}{3}$ cubic chi.

The Chu Tong (芻童)

術：倍上表以下表從之，以上廣乘之；又倍下表以上表從之，以下廣乘之。并之，以高乘之，六而一。

Method: Double the top length and add the bottom length; multiply by the top width. Separately, double the bottom length and add the top length; multiply by the bottom width. Add these two results, multiply by height, and divide by 6.

PROBLEM

芻童下廣三丈，表四丈，上廣二丈，表三丈，高三丈。問積？答曰：二萬六千五百尺。 A *chutong* has bottom width 3 zhang, length 4 zhang, top width 2 zhang, length 3 zhang, and height 3 zhang. **Answer:** 26,500 cubic chi.

SOLUTION (草曰):

$(2 \times 30 + 40) \times 20 + (2 \times 40 + 30) \times 30 = 2000 + 3300 = 5300$ 。
積 = $\frac{5300 \times 30}{6} = 26500$ 立方尺。

$(2 \times 30 + 40) \times 20 + (2 \times 40 + 30) \times 30 = 2000 + 3300 = 5300$.
Volume = $\frac{5300 \times 30}{6} = 26,500$ cubic chi.

10 Chapter 7: Equitable Distribution / 均輸

均輸章論各地區之賦役與運輸成本之公平分配。凡二十八題。

The Equitable Distribution chapter deals with fair allocation of tax burdens and transportation costs across different regions. It originally contained 28 problems.

10.1 The Equitable Distribution Method / 均輸法

術：以一斛之運費與各縣距離相乘，加粟價，得各縣之比率。以戶數為衰，副并為法。以賦粟乘未并之衰為實。實如法而一。

Method: Multiply the distance and transport cost per unit for each county. Add the grain price to obtain the proportional rate. Simplify the county household numbers to form the distribution rates. Set out the rates, combine as the divisor. Multiply the total grain tax by each uncombined rate for the dividends. Divide each by the divisor.

PROBLEM

五縣均賦粟一萬石，每車載二十五石，行道一里，出雇錢一文。各縣到輸所遠近不等，粟價高下，欲令縣勞費相等。Five counties are to share equally the tax burden of 10,000 dan of millet. Each cart carries 25 dan, and for each li traveled, the transport cost is 1 qian. The counties differ in distance from the delivery point and in grain prices. How much does each county contribute?

SOLUTION (草曰):

每石每里運費 = $\frac{1}{25}$ 錢。

各縣率：

甲： $20 + 0 = 20$ ；乙： $10 + \frac{200}{25} = 18$ ；

丙： $12 + \frac{150}{25} = 18$ ；丁： $17 + \frac{250}{25} = 27$ ；

戊： $13 + \frac{150}{25} = 19$ 。

按戶數 × 率分配各縣賦額。

Transport cost per dan per li = $\frac{1}{25}$ qian.

Rates: A: $20 + 0 = 20$; B: $10 + \frac{200}{25} = 18$;

C: $12 + \frac{150}{25} = 18$; D: $17 + \frac{250}{25} = 27$;

E: $13 + \frac{150}{25} = 19$.

Distribution proportional to households × rate.

10.2 The Pursuit Problem / 相遇問題

PROBLEM

甲發長安，五日至齊，乙發齊，七日至長安。今乙先行二日。問：幾日相逢？答曰：二日，十二分日之一。Person A departs from Chang'an and reaches Qi in 5 days. Person B departs from Qi and reaches Chang'an in 7 days. Now B has already departed 2 days ahead. Answer: $2\frac{1}{12}$ days.

SOLUTION (草曰):

甲速 = $\frac{1}{5}$ (每日行程)；乙速 = $\frac{1}{7}$ 。

乙先行二日，已行 $\frac{2}{7}$ 。餘程 = $\frac{5}{7}$ 。

合速 = $\frac{1}{5} + \frac{1}{7} = \frac{12}{35}$ 。

時 = $\frac{5/7}{12/35} = \frac{175}{84} = 2\frac{1}{12}$ 日。

A's speed = $\frac{1}{5}$ (of the distance per day). B's speed = $\frac{1}{7}$.

In 2 days, B covers $\frac{2}{7}$ of the distance. Remaining = $\frac{5}{7}$.

Combined speed = $\frac{1}{5} + \frac{1}{7} = \frac{12}{35}$.

Time = $\frac{5/7}{12/35} = \frac{175}{84} = 2\frac{1}{12}$ days.

10.3 The Gourd and Melon / 瓜瓠相逢

PROBLEM

垣高九尺，瓜生其上，蔓日長七寸；瓠生其下，蔓日長一尺。問：幾日相逢？答曰：相逢於五日，十七分日之五。A wall is 9 chi high. A gourd grows from the top, extending 7 cun per day. A bottle-gourd grows from the bottom, extending 1 chi per day. Answer: They meet after $5\frac{5}{17}$ days.

SOLUTION (草曰):

每日共長 = $7 + 10 = 17$ 寸。

時 = $\frac{90}{17} = 5\frac{5}{17}$ 日。

瓜長 = $\frac{35}{17} \times 7 = \frac{245}{17} = 14\frac{7}{17}$ 寸。

瓠長 = $\frac{35}{17} \times 10 = \frac{350}{17} = 20\frac{10}{17}$ 寸。

Total daily growth = 7 cun + 10 cun = 17 cun per day.

Time = $\frac{90}{17} = 5\frac{5}{17}$ days.

Gourd vine: $\frac{35}{17} \times 7 = \frac{245}{17} = 14\frac{7}{17}$ cun.

Bottle-gourd: $\frac{35}{17} \times 10 = \frac{350}{17} = 20\frac{10}{17}$ cun.

11 Chapter 8: Excess and Deficit / 盈不足

盈不足章介紹盈不足術—即雙假設法，為解線性方程與近似問題之強大方法。凡二十題。

The Excess and Deficit chapter introduces the “method of double false position”—a powerful technique for solving linear equations and approximation problems. It originally contained 20 problems.

11.1 The Excess and Deficit Method / 盈不足法

術：置所出率，盈、不足各居其下。維乘所出率，并以為實。并盈、不足為法。實如法而一。買物者，以所出率以少減多，餘以約法、實。實為物價，法為人數。

Method: Set out the proposed rates, placing the excess or deficit below each. Cross-multiply the rates by the excess/deficit and sum for the dividend. Sum the excess and deficit for the divisor. Divide the dividend by the divisor. For purchasing problems: subtract the smaller rate from the larger for the simplified divisor. The dividend gives the price, the divisor gives the number of people.

PROBLEM

共買物，人出八文，盈三文，人出七文，不足四文。問數、物價各幾何？答曰：七人。物價五十三。Several people jointly buy an item. If each pays 8 qian, there is an excess of 3 qian. If each pays 7 qian, there is a deficit of 4 qian. **Answer:** 7 people; price 53 qian.

SOLUTION (草曰):

維乘： $8 \times 4 + 7 \times 3 = 32 + 21 = 53$ 。
法 = $3 + 4 = 7$ 。
物價 = $\frac{53}{1} = 53$ 文。
人數 = $\frac{3+4}{8-7} = 7$ 。

$Dividend = 8 \times 4 + 7 \times 3 = 32 + 21 = 53$.
 $Divisor = 3 + 4 = 7$.
 $Price = \frac{53}{1} = 53 \text{ qian}$.
 $Number \text{ of people} = \frac{3+4}{8-7} = 7$.

PROBLEM

共買雞，各出九，盈十一，各出六，不足十六。問人數、雞價各幾何？答曰：九人，雞價七十。Several people jointly buy a chicken. If each pays 9 qian, there is an excess of 11 qian. If each pays 6 qian, there is a deficit of 16 qian. **Answer:** 9 people; price 70 qian.

SOLUTION (草曰):

人數 = $\frac{11+16}{9-6} = \frac{27}{3} = 9$ 。
雞價 = $9 \times 9 - 11 = 81 - 11 = 70$ 文。

$Number \text{ of people} = \frac{11+16}{9-6} = \frac{27}{3} = 9$.
 $Price = 9 \times 9 - 11 = 81 - 11 = 70 \text{ qian}$.

11.2 Excess and Deficit with Exact Amount / 盈朒適足

PROBLEM

共買犬，人出五不足九十文，人出五十文適足。問人數、犬價各幾何？答曰：二人，犬價一百。Several people jointly buy a dog. If each pays 5 qian, there is a deficit of 90 qian. If each pays 50 qian, it is exactly sufficient. **Answer:** 2 people; price 100 qian.

SOLUTION (草曰):

設人數為 n ，犬價為 P 。
 $5n + 90 = 50n \Rightarrow 45n = 90 \Rightarrow n = 2$ 。
 $P = 50 \times 2 = 100$ 文。

Let $n = \text{number of people}$, $P = \text{price}$.
 $5n + 90 = 50n \Rightarrow 45n = 90 \Rightarrow n = 2$.
 $P = 50 \times 2 = 100 \text{ qian}$.

12 Chapter 9: Systems of Linear Equations / 方程

方程章為中國數學中最卓越者之一。其以算籌排列係數，用類似高斯消去法之技術求解線性方程組。凡十八題。

The Equations chapter is one of the most remarkable in Chinese mathematics. It presents methods for solving systems of linear equations using matrix-like arrays of counting rods—a technique equivalent to modern Gaussian elimination. It originally contained 18 problems.

12.1 The Fang Cheng Method / 方程法

術：置行數相乘，以少減多，反覆相消，直至一未知數孤立。錢數為實，物數為法。

Method: Set up the coefficients in an array. For the quantities sought, cross-multiply adjacent rows and repeatedly subtract the smaller from the larger until one variable is isolated. The money amounts form the dividend, the quantities form the divisor.

PROBLEM Three Grains / 三禾

上禾三秉，中禾二秉，下禾一秉，共實三十九斗。上禾二秉，中禾三秉，下禾一秉，共實三十四斗。上禾一秉，中禾二秉，下禾三秉，共實二十六斗。問：上、中、下禾一秉各幾何？ 3 bundles of top-grade grain, 2 bundles of medium-grade, and 1 bundle of low-grade yield 39 dou. 2 bundles of top-grade, 3 bundles of medium-grade, and 1 bundle of low-grade yield 34 dou. 1 bundle of top-grade, 2 bundles of medium-grade, and 3 bundles of low-grade yield 26 dou. **Answer:** Top $9\frac{1}{4}$; Medium $4\frac{1}{4}$; Low $2\frac{3}{4}$ dou per bundle.

SOLUTION (草曰):

方程組：

$$3x + 2y + z = 39$$

$$2x + 3y + z = 34$$

$$x + 2y + 3z = 26$$

用方程術（高斯消去）：

由一、二式得： $x - y = 5$ 。

由二、三式得： $5x - 7y = 1$ 。

解之： $x = 9\frac{1}{4}$, $y = 4\frac{1}{4}$, $z = 2\frac{3}{4}$ 。

System:

$$3x + 2y + z = 39$$

$$2x + 3y + z = 34$$

$$x + 2y + 3z = 26$$

Using the fangcheng method (Gaussian elimination):

From equations (1) and (2): $x - y = 5$.

From equations (2) and (3): $5x - 7y = 1$.

Solving: $x = 9\frac{1}{4}$, $y = 4\frac{1}{4}$, $z = 2\frac{3}{4}$.

12.2 Cattle and Sheep / 牛羊直金

PROBLEM

五牛二羊，直金十兩，二牛五羊，直金八兩。問：牛、羊價各幾何？ 答曰：一牛直金一兩二十一分兩之十三；一羊直金二十一分兩之二十。 5 oxen and 2 sheep are worth 10 liang of gold. 2 oxen and 5 sheep are worth 8 liang of gold. **Answer:** Ox $1\frac{13}{21}$ liang; Sheep $\frac{20}{21}$ liang.

SOLUTION (草曰):

$$5o + 2s = 10, 2o + 5s = 8.$$

$$\text{維乘: } 25o + 10s = 50, 4o + 10s = 16.$$

$$\text{相減: } 21o = 34, \text{ 故 } o = \frac{34}{21} = 1\frac{13}{21} \text{ 兩.}$$

$$2s = 10 - 5 \times \frac{34}{21} = \frac{40}{21}, \text{ 故 } s = \frac{20}{21} \text{ 兩.}$$

$$\text{System: } 5o + 2s = 10, 2o + 5s = 8.$$

$$\text{Cross-multiply: } 25o + 10s = 50, 4o + 10s = 16.$$

$$\text{Subtract: } 21o = 34, \text{ so } o = \frac{34}{21} = 1\frac{13}{21} \text{ liang.}$$

$$2s = 10 - 5 \times \frac{34}{21} = \frac{40}{21}, \text{ so } s = \frac{20}{21} \text{ liang.}$$

13 Chapter 10: Right Triangles / 勾股

勾股章應用勾股定理理解實際問題。凡三十八題。

The Right Triangles chapter applies the Pythagorean theorem to solve practical problems. It originally contained 38 problems.

13.1 Pythagorean Theorem / 勾股定理

求弦術：勾股各自乘，并而開方除之，即弦。

求勾術：弦自乘，減股自乘，餘開方除之，即勾。

Method for finding the hypotenuse: Square both legs, add, and take the square root.

Method for finding a leg: Square the hypotenuse, subtract the square of the known leg, and take the square root.

PROBLEM

勾八尺，股十五尺。問為弦幾何？答曰：十七尺。 A right triangle has legs (gou) of 8 chi and (gu) of 15 chi. What is the hypotenuse (xian)? **Answer:** 17 chi.

SOLUTION (草曰):

弦 = $\sqrt{8^2 + 15^2} = \sqrt{64 + 225} = \sqrt{289} = 17$ 尺。

$xian = \sqrt{8^2 + 15^2} = \sqrt{64 + 225} = \sqrt{289} = 17$ chi.

13.2 The Vine and the Tree / 葛纏木

PROBLEM

木長二丈，圍之三尺，葛生其下，纏木七周，上與木齊。問葛長幾何？答曰：二丈九尺。 A tree is 2 zhang tall with a circumference of 3 chi. A vine grows at the base, winds around the tree 7 times, and reaches the top. **Answer:** 2 zhang 9 chi.

SOLUTION (草曰):

展開圓柱：葛為直角三角形之斜邊。

一邊 = 樹高 = 20 尺；另一邊 = $7 \times 3 = 21$ 尺。

葛長 = $\sqrt{20^2 + 21^2} = \sqrt{841} = 29$ 尺 = 二丈九尺。

Unwrapping the cylinder: the vine forms the hypotenuse of a right triangle with:

One leg = height of tree = 20 chi; Other leg = $7 \times 3 = 21$ chi.

Vine length = $\sqrt{20^2 + 21^2} = \sqrt{841} = 29$ chi = 2 zhang 9 chi.

13.3 The Reed in the Pond / 葭生中央

PROBLEM

池方一丈，正中有葭，出水面一尺，引葭至岸，與水面適平。問水深？答曰：水深一丈二尺。 A square pond has side 1 zhang. A reed grows at its center, protruding 1 chi above the water. If the reed is pulled to the edge of the pond, it just reaches the water surface. **Answer:** Water depth 1 zhang 2 chi.

SOLUTION (草曰):

設水深 d 。葭長 = $d + 1$ 。

引至岸：水平距 = 5 尺（半池邊）。

由勾股定理： $(d + 1)^2 = d^2 + 5^2 = d^2 + 25$ 。

$d^2 + 2d + 1 = d^2 + 25 \Rightarrow 2d = 24 \Rightarrow d = 12$ 尺。

Let d = water depth. Reed length = $d + 1$.

When pulled to edge: horizontal distance = 5 chi (half the pond side).

By Pythagorean theorem: $(d + 1)^2 = d^2 + 5^2 = d^2 + 25$.

$d^2 + 2d + 1 = d^2 + 25 \Rightarrow 2d = 24 \Rightarrow d = 12$ chi.

13.4 The Broken Bamboo / 折竹抵地

PROBLEM

竹高一丈，折梢拄地，去根三尺，問折處高幾何？答曰：四尺，二十分尺之十一。 A bamboo is 1 zhang tall. It breaks and the top touches the ground 3 chi from the root. **Answer:** 4 $\frac{11}{20}$ chi.

SOLUTION (草曰):

設折處高 h 。折下部分 = $10 - h$ 。

由勾股定理： $(10 - h)^2 = h^2 + 3^2 = h^2 + 9$ 。

$100 - 20h + h^2 = h^2 + 9 \Rightarrow 20h = 91 \Rightarrow h = \frac{91}{20} = 4 \frac{11}{20}$ 尺。

Let h = height of break. The fallen part = $10 - h$.

By Pythagorean theorem: $(10 - h)^2 = h^2 + 3^2 = h^2 + 9$.

$100 - 20h + h^2 = h^2 + 9 \Rightarrow 20h = 91 \Rightarrow h = \frac{91}{20} = 4 \frac{11}{20}$ chi.

13.5 Circle Inscribed in a Right Triangle / 勾股容圓

PROBLEM

勾八步，股十五步，問勾中容圓徑幾何？答曰：六步。 A right triangle has legs (gou) 8 bu and (gu) 15 bu. What is the diameter of the inscribed circle? **Answer:** 6 bu.

SOLUTION (草曰):

$$\text{弦} = \sqrt{8^2 + 15^2} = 17 \text{ 步。}$$

$$\text{積} = \frac{1}{2} \times 8 \times 15 = 60 \text{ 平方步。}$$

$$\text{半周} = \frac{8+15+17}{2} = 20 \text{ 步。}$$

$$\text{內切圓半徑} = \frac{\text{積}}{\text{半周}} = \frac{60}{20} = 3 \text{ 步。}$$

$$\text{直徑} = 2 \times 3 = 6 \text{ 步。}$$

$$\text{Hypotenuse} = \sqrt{8^2 + 15^2} = 17 \text{ bu.}$$

$$\text{Area} = \frac{1}{2} \times 8 \times 15 = 60 \text{ square bu.}$$

$$\text{Semi-perimeter} = \frac{8+15+17}{2} = 20 \text{ bu.}$$

$$\text{Inradius} = \frac{\text{Area}}{\text{semi-perimeter}} = \frac{60}{20} = 3 \text{ bu.}$$

$$\text{Diameter} = 2 \times 3 = 6 \text{ bu.}$$

14 Square and Cube Root Extraction / 開方術

楊輝保存並解釋了賈憲的開方方法，此為宋代計算數學之巔峰。

Yang Hui preserved and explained Jia Xian's methods for root extraction, which represented the pinnacle of Song Dynasty computational mathematics.

14.1 Jia Xian's "Unlocking" Square Root Method / 賈憲釋鎖開平方法

術：一、置積為實。二、借一算步之，超一等。三、商所得。四、以商乘下法為方法，而以除。五、除已，倍方法為廉法。六、復置下法，退位，復商。七、以商乘廉法，命實除之。

Method: (1) Set the number whose root is sought as the dividend (實). (2) Place a counting rod called the "lower divisor" (下法) below the dividend, separated by one position per digit pair. (3) Estimate the first digit of the root and place it above. (4) Multiply this digit by the lower divisor to form the "square divisor" (方法). (5) Subtract from the dividend. (6) Double the square divisor to form the "flank divisor" (廉法). (7) Shift positions and repeat for subsequent digits.

PROBLEM

積五萬五千二百二十五步，問為方幾何？答曰：二百三十五步。An area of 55,225 square bu. What is the side of the square? **Answer:** 235 bu.

SOLUTION (草曰):

$\sqrt{55225} = 235$ 步。

步驟：

一、分節：5 | 52 | 25。

二、首商： $2^2 = 4 \leq 5$ ，首商為二。

三、餘： $5 - 4 = 1$ ，下推 52 得 152。

四、倍法： $2 \times 2 = 4$ 。求 d 使 $(40 + d) \times d \leq 152$ 。 $d = 3$ ：
 $43 \times 3 = 129$ 。

五、餘： $152 - 129 = 23$ ，下推 25 得 2325。

六、倍法： $23 \times 2 = 46$ 。求 d 使 $(460 + d) \times d \leq 2325$ 。 $d = 5$ ：
 $465 \times 5 = 2325$ 。

七、餘零。故 $\sqrt{55225} = 235$ 。

$\sqrt{55225} = 235$ bu.

Step-by-step:

(1) Divide into groups: 5 | 52 | 25.

(2) First digit: $2^2 = 4 \leq 5$, so first digit is 2.

(3) Remainder: $5 - 4 = 1$. Bring down 52: 152.

(4) Double the current result: $2 \times 2 = 4$. Find d such that $(40 + d) \times d \leq 152$. $d = 3$: $43 \times 3 = 129$.

(5) Remainder: $152 - 129 = 23$. Bring down 25: 2325.

(6) Double current result: $23 \times 2 = 46$. Find d such that $(460 + d) \times d \leq 2325$. $d = 5$: $465 \times 5 = 2325$.

(7) Remainder: 0. So $\sqrt{55225} = 235$.

14.2 Cube Root Extraction / 開立方術

術：類似開平方，但有三級除數：方法、廉法、隅法。

Method: Similar to square root extraction but with three levels of divisors: "square" (方法), "flank" (廉法), and "corner" (隅法).

PROBLEM

積一百八十六萬八百六十七尺，問為立方幾何？答曰：一百二十三尺。A volume of 1,860,867 cubic chi. What is the side of the cube? **Answer:** 123 chi.

SOLUTION (草曰):

$\sqrt[3]{1860867} = 123$ 尺。

驗算： $123^3 = 123 \times 123 \times 123 = 15129 \times 123 = 1860867$ 。✓

$\sqrt[3]{1860867} = 123$ chi.

Verification: $123^3 = 123 \times 123 \times 123 = 15129 \times 123 = 1,860,867$. ✓

Original Chinese (原文)

English Translation (英譯)

15 Classification of Methods / 算類

楊輝將二百四十六題按所用之數學方法系統分類。他發現不同章節之題往往運用相同之基本技巧。

Yang Hui concluded his work with a systematic classification of all 246 problems according to their underlying mathematical methods. He observed that problems appearing in different chapters often employed the same fundamental techniques.

15.1 Method Cross-Reference Table / 方法交叉分類表

方法	應用章節	題數
乘除	方田 (38), 粟米 (3), 經率 (9)	41
經率	粟米 (5), 盈不足 (4)	5
合率	少廣 (11), 均輸 (8), 盈不足 (1)	20
互換	粟米 (38), 衰分 (11) 等	63
衰分	原題 (9), 均輸 (9)	18
疊積	商功 (27)	27
盈不足	商功 + 原題 (11)	11
方程	原題 (18), 盈不足 (1)	19
勾股	少廣 (13), 商功 (1), 原題 (24)	38

Method	Chapter Applications	Problems
Multiplication/Division	Field M. (38), Grain (3), Direct R. (9)	41
Direct Rate	Grain (5), Excess/Def. (4)	5
Combined Rates	Short W. (11), Equit. (8), E/D (1)	20
Proportional Exch.	Grain (38), Prop. D. (11), etc.	63
Prop. Distribution	Original (9), Equit. (9)	18
Stacked Volumes	Comm. (27)	27
Excess/Deficit	Comm. + Orig. (11)	11
Equations	Original (18), E/D (1)	19
Right Triangles	S.W. (13), Comm. (1), Orig. (24)	38

15.2 Yang Hui's Critical Observations / 楊輝批判性論述

楊輝對傳世文本提出數項重要批判：

一、部分粟米章之比例交換題及少廣章之開方題冗餘或位置不當。

二、部分題目顯為後人增補，非《九章》原文。

三、部分古法已廢棄不用，需予恢復。

Yang Hui made several important critical observations about the transmitted text:

(1) Some problems in the Grain chapter on proportional exchange and in the Short Width chapter on root extraction were redundant or misplaced.

(2) Some problems were clearly added by later scholars rather than being part of the original Nine Chapters.

(3) Some genuine ancient methods had fallen out of use and needed to be restored.

楊輝竊見九章舊本，作立題法遺闕。古序云：靖康以來，罕有舊本，間有存者，狃於末習，不循本意，以至真術淹廢，偽本滋典。

“I have examined old copies of the Nine Chapters and noted missing problems and methods. The ancient preface states: Since the Jingkang period, old copies have been rare, and those occasionally surviving have been corrupted by inferior practices, not following the original intent, so that genuine methods have been submerged and abandoned, while corrupt copies have proliferated.”

Original Chinese (原文)	English Translation (英譯)
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16 Conclusion / 結論

楊輝《詳解九章算法》為中國數學史之里程碑。其貢獻包括：

一、**保存賈憲著作**：經楊輝引述，後人得知賈憲之二項式三角、開方方法及代數貢獻—否則此等著作將完全失傳。

二、**系統闡述**：楊輝之四部結構（問、答、術、草）成為數學著述之典範。

三、**批判性學術**：其按方法而非按主題分類問題，揭示不同應用領域中數學技巧之統一性。

四、**教學創新**：楊輝明言其著述為助「不得其門而入」之學子。

開方作法本源圖（楊輝三角）已成為數學中最著名之圖形之一，出現於全球教科書中。其在西方被稱為「帕斯卡三角」—以布萊茲·帕斯卡（一六二三至一六六二年）命名，而帕斯卡晚於賈憲約六百年、晚於楊輝約四百年—此乃數學知識跨文化獨立發展之明證，亦提醒我們中國數學家對現代數學基礎之重要貢獻。

Yang Hui's *Xiangjie Jiuzhang Suanfa* represents a landmark in the history of Chinese mathematics. Its contributions include:

(1) **Preservation of Jia Xian's work**: Through Yang Hui's citations, we know of Jia Xian's binomial triangle, his root extraction methods, and his contributions to algebra—works that would otherwise be completely lost.

(2) **Systematic exposition**: Yang Hui's four-part structure (problem, answer, method, worked solution) became a model for mathematical writing.

(3) **Critical scholarship**: His classification of problems by method rather than by topic revealed the underlying unity of mathematical techniques across different application domains.

(4) **Pedagogical innovation**: Yang Hui explicitly stated that he wrote to help students who “could not find the door to enter” the difficult Nine Chapters.

The “Method Source for Square Root Extraction” (Yang Hui's Triangle) has become one of the most celebrated images in mathematics, appearing in textbooks worldwide. That it should be known in the West as “Pascal's Triangle”—named after Blaise Pascal (1623–1662), who lived some 600 years after Jia Xian and 400 years after Yang Hui—is a testament to the independent development of mathematical knowledge across cultures, and a reminder of the importance of recognizing the contributions of Chinese mathematicians to the foundations of modern mathematics.

「數學為眾學之基。」
— 楊輝《詳解九章算法》

“Mathematics is the foundation of all learning.”
— Yang Hui, *Xiangjie Jiuzhang Suanfa*

詳解九章算法† *Xiangjie Jiuzhang Suanfa*

(二) *Part II*

Bilingual Edition 雙語版

Chinese Original with English Translation 中文原文附英譯

宋 楊輝 撰

Written by Yang Hui of the Song Dynasty

(c. 1238–1298 CE)

Bilingual Typeset Edition 雙語排印版

May 27, 2026

Left column: Original Chinese text 左欄: 中文原文

Right column: English translation 右欄: 英譯

This volume covers chapters on Fair Levies (均輸), Excess and Deficit (盈不足),
Rectangular Arrays (方程), Right Triangles (勾股), and Construction (商功).

Contents

Introduction 導言	2
1 Collation Notes 詳解九章算法札記	3
2 Chapter on Fair Levies 均輸第六	4
2.1 Problem: Equalizing Transportation Costs 均輸粟	4
2.2 Problem: The Good and Slow Walkers 善行者與拙行者	4
2.3 Problem: The Hare and the Hound 犬追兔	5
2.4 Problem: Distance Calculation 空車重車	5
2.5 Problem: Meeting of Duck and Goose 鳧雁相逢	5
3 Chapter on Excess and Deficit 盈不足第七	6
3.1 The Method of Excess and Deficit 盈不足法	6
3.2 Problem: Buying a Chicken 共買雞	6
3.3 Problem: Buying Gold 共買金	7
3.4 Problem: The Good and Inferior Fields 善田與惡田	7
3.5 Problem: Large and Small Vessels 大器與小器	7
3.6 Problem: The Horses of Liang 良馬與駑馬	8
4 Chapter on Rectangular Arrays 方程第八	9
4.1 The Method of Rectangular Arrays 方程法	9
4.2 Problem: Three Grades of Grain 上禾、中禾、下禾	10
4.3 Problem: Five Families Sharing a Well 五家共井	11
4.4 Problem: Gold and Silver Weights 金與銀	11
4.5 Problem: Buying Oxen, Sheep, and Pigs 牛、羊、豕	12
5 Chapter on Right Triangles 勾股第九	13
5.1 Key Formulas 勾股求弦	13
5.2 Problem: Finding the Hypotenuse 勾三股四弦五	13
5.3 Problem: Finding the Height 弦十七步	14
5.4 Problem: The Reed in the Pond 葭生池中	14
5.5 Problem: The Door and the Pole 戶高與廣	14
5.6 Problem: The Square Inscribed in a Right Triangle 勾股容方	15
5.7 Problem: Measuring the Square City 邑方測量	15
6 Chapter on Construction 商功	16
6.1 Volume Formulas 求積公式	16
6.2 Problem: Volume of a City Wall 城垣積	16
6.3 Problem: Square Tower 方亭臺	17
6.4 Problem: Round Tower 圓亭臺	17
6.5 Problem: Grain Pile on the Ground 委粟平地	17
7 Yang Hui's Triangle 開方作法本源	18
7.1 Application to Root Extraction 開方之應用	19
Appendix 附錄：九章纂類	20

Introduction 導言

詳解九章算法 (*Xiangjie Jiuzhang Suanfa*, 「Detailed Analysis of the Mathematical Methods in the Nine Chapters」) 為南宋 (1127–1279 年) 最重要之數學著作之一。此書由楊輝 (約 1238–1298 年) 撰於 1261 年, 共十二卷, 詳解《九章算術》八十題 (原為九十七題)。

此卷 (第二部分) 包含楊輝著作之後數章:

- 均輸: 均輸——關於賦稅、勞役分配及運輸之問題
- 盈不足: 盈不足——雙假設位法
- 方程: 方程——線性方程組
- 勾股: 勾股——中國勾股定理及其應用
- 商功: 商功——各種幾何體之體積計算

楊輝之著作以保存現已失傳之早期數學方法而著稱, 包括賈憲 (賈憲) 之「增乘開方法」及劉益 (劉益) 對四次方程之貢獻。

Xiangjie Jiuzhang Suanfa (“Detailed Analysis of the Mathematical Methods in the Nine Chapters”) is one of the most important mathematical works of the Southern Song dynasty (1127–1279 CE). Written by Yang Hui (c. 1238–1298) and published in 1261 CE, this 12-volume work provides detailed explanations of 80 selected problems from the ancient *Jiuzhang Suanshu* (Nine Chapters on the Mathematical Art).

This volume (Part II) contains the latter chapters of Yang Hui’s work:

- **Junshu** (均輸): Fair Levies — problems on taxation, labor distribution, and transportation
- **Ying Buzu** (盈不足): Excess and Deficit — the method of double false position
- **Fangcheng** (方程): Rectangular Arrays — systems of linear equations
- **Gougu** (勾股): Right Triangles — the Chinese Pythagorean theorem and its applications
- **Shanggong** (商功): Construction — volumes of various geometric solids

Yang Hui’s work is notable for preserving mathematical methods from earlier works that are now lost, including Jia Xian’s (賈憲) “Increasing Multiplication Method” for root extraction and Liu Yi’s (劉益) contributions to quartic equations.

1 Collation Notes 詳解九章算法札記

詳解九章算法者，宋錢塘楊輝謙光取古九章算經，抄錄經注原文於前，而以其所撰圖解釋注比類圖說驗辭各條之後者也。末附以九章纂類，則以當時俗傳算術為細而分析九章題問以類相從焉。

據自序，詳解八十題，乃九十七題，綿十二卷，今不分卷，蓋非原書，故其中抄錄經注亦多不循舊次。而世無傳本，無從校核。儀徵阮誼國收藏算書最富，而正續疇人傳俱云未見，餘可知矣。

全案九章為算經之首，諸家立術皆自此出。而世傳永樂大典及孔氏微波榭二本，均不免脫誤。鐘祥李尚書細草圖說多所改正，方可卒讀，而往往與此書暗合，則此書誠可貴也。且其圖靈中所列名目亦足與宋人算書互證。

The *Xiangjie Jiuzhang Suanfa* was written by Yang Hui (Qianguang) of Qiantang during the Song dynasty. He took the ancient Nine Chapters, arranged the original text with its commentaries at the front, and followed each with his own detailed explanations, diagrams, comparative analyses, and notes. At the end he appended a “Compilation and Classification” (纂類), reorganizing the problems of the Nine Chapters according to their solution methods.

According to the original preface, the detailed analysis covers 80 problems (originally 97 problems), in 12 volumes. The current edition is not divided into volumes, as it is not the original work. The transmitted text does not follow the original order. No complete copy has survived, making collation impossible. Ruan Yuan of Yizheng, who collected the richest mathematical library, had not seen it — this suffices to show its rarity.

The Nine Chapters is the foremost mathematical classic, and all schools of mathematics derive their methods from it. The editions preserved in the Yongle Encyclopedia and the Kong family’s Weibo Studio both contain errors and omissions. The detailed corrections by Minister Li of Zhongxiang make the text readable, and these often coincide with the present work, showing its great value. The diagrams and terminology listed therein also serve as mutual corroboration with other Song dynasty mathematical texts.

2 Chapter on Fair Levies 均輸第六

均輸章處理賦稅、勞役分配及運輸之公平問題。其將前數章之比例方法推廣至涉及多種比率、距離及工作分配之更複雜情境。

均輸之「均」意為「平均」或「公平」，「輸」意為「輸送」或「運輸」。此方法確保各郡縣按人口、距離及物價分擔公共工程與運輸之費用。

The *Junshu* (“Fair Levies”) chapter deals with problems of equitable taxation, labor distribution, and transportation. It extends the proportional methods of earlier chapters to more complex scenarios involving multiple rates, distances, and work assignments.

The term *jun* (均) means “equal” or “fair,” while *shu* (輸) means “to transport” or “to convey.” This method ensures that counties share the costs of public works and transportation according to their population, distance, and commodity prices.

2.1 Problem: Equalizing Transportation Costs 均輸粟

問：均輸粟，以一車二十五斛。加一斛粟價加一斛之價，凡輸粟取僦錢。問如何公平分配？

術：以比例分配法。列各郡之比率，乘以運輸量，除以總數，得各郡之公平份額。

若甲郡出 a 單位，比率 r_A ；乙郡出 b 單位，比率 r_B ，則公平分配為：

$$\text{甲份} = \frac{a \cdot r_A}{a \cdot r_A + b \cdot r_B} \times \text{總額}$$

Problem: Transportation of grain: one cart carries 25 *hu* (斛). When the price of one additional *hu* of grain is added, the total transportation cost is calculated. How should the costs be fairly distributed?

Method: Use the rule of proportional distribution. Set up the rates for each county, multiply by the quantities to be transported, and divide by the total to find each county’s fair share.

Mathematically: If county A contributes a units at rate r_A , county B contributes b units at rate r_B , then:

$$\text{Share}_A = \frac{a \cdot r_A}{a \cdot r_A + b \cdot r_B} \times \text{Total}$$

2.2 Problem: The Good and Slow Walkers 善行者與拙行者

問：善行者一百步，拙行者六十步。今拙者先行一百步，問善行者幾何步追及？

術：善行者每行百步，追及拙行者四十步。以拙者先行百步乘善行者百步為實，以四十步為法。實如法而一。

$$\text{所需步數} = \frac{100 \times 100}{100 - 60} = \frac{10000}{40} = 250$$

此即相對速率問題之經典範例。

Problem: A fast walker takes 100 paces while a slow walker takes 60 paces. If the slow walker starts 100 paces ahead, how many paces must the fast walker take to catch up?

Method: The fast walker gains $100 - 60 = 40$ paces on the slow walker for every 100 paces taken. To make up a deficit of 100 paces, multiply the head start (100) by the fast walker’s pace (100) to obtain the dividend. Take the difference (40) as the divisor. Divide the dividend by the divisor:

$$\text{Paces needed} = \frac{100 \times 100}{100 - 60} = \frac{10000}{40} = 250$$

This is a classic example of a relative speed problem.

2.3 Problem: The Hare and the Hound 犬追兔

問：犬追兔，兔先一百步。犬發，追一百五十步，不及一十步而止。問犬不止，更追幾何步及之？

答：一百七步七分步之一。

術：犬行百五十步，不及十步。令犬再行，所追步數以比例求之。犬追兔之速差，以不及步數乘之，得再行步數。

$$\text{再行步數} = \frac{150 \times 10}{40 + 10} = \frac{1500}{50} = 30$$

楊輝注：此術亦可用盈不足法求之。

Problem: A hound chases a hare. The hare has a 100-pace head start. The hound pursues for 150 paces but is still 10 paces behind. If the hound continues, how many more paces until it catches the hare?

Answer: $107\frac{1}{7}$ paces.

Method: In 150 paces the hound reduces the gap by 40 paces but is still 10 paces behind. Using proportion: multiply the deficit (10) by the paces already run (150), and divide by the rate of closure:

$$\text{Additional paces} = \frac{150 \times 10}{40 + 10} = \frac{1500}{50} = 30$$

Yang Hui's commentary notes: This method can also be solved by the excess-deficit method.

2.4 Problem: Distance Calculation 空車重車

問：空車日行七十里，重車日行五十里。今載粟至倉，五返，問遠幾何？

答：四十八里十八分里之十一。

術：設距離為 d 。一往一返所需日數為 $\frac{d}{50} + \frac{d}{70}$ 。五返為指定日數，解 d 即可。

此問題涉及調和平均數之概念。

Problem: An empty cart travels 70 *li* (里) per day; a loaded cart travels 50 *li* per day. Now grain is transported to the granary, making 5 round trips. How far is the granary?

Answer: $48\frac{11}{18}$ *li*.

Method: Let d be the distance. One round trip takes $\frac{d}{50} + \frac{d}{70}$ days. Five round trips take one day (or a specified time). Solve for d .

This problem involves the concept of the harmonic mean.

2.5 Problem: Meeting of Duck and Goose 鳧雁相逢

問：鳧起南海，七日至北海；雁起北海，九日至南海。今鳧雁俱起，問何日相逢？

答：三日十六分日之十五。

術：鳧一日行全程 $\frac{1}{7}$ ，雁一日行全程 $\frac{1}{9}$ 。合行 $\frac{1}{7} + \frac{1}{9} = \frac{16}{63}$ 。相逢之日：

$$\frac{1}{\frac{1}{7} + \frac{1}{9}} = \frac{63}{16} = 3\frac{15}{16} \text{ 日}$$

此為最早之「相遇問題」範例之一。

Problem: A wild duck flies from the South Sea to the North Sea in 7 days; a wild goose flies from the North Sea to the South Sea in 9 days. If they start simultaneously, on what day will they meet?

Answer: $3\frac{15}{16}$ days.

Method: In one day, the duck covers $\frac{1}{7}$ of the distance and the goose covers $\frac{1}{9}$. Together they cover $\frac{1}{7} + \frac{1}{9} = \frac{16}{63}$ per day. They meet after:

$$\frac{1}{\frac{1}{7} + \frac{1}{9}} = \frac{63}{16} = 3\frac{15}{16} \text{ days}$$

This is one of the earliest examples of a “meeting problem.”

3 Chapter on Excess and Deficit 盈不足第七

盈不足章介紹雙假設位法（盈不足術），一種通過兩次猜測並利用誤差求得正確解之技術。此法可解線性及某些非線性問題。

「盈」意為「過剩」，「不足」意為「短缺」。此二詞分別代表兩次嘗試中超出或不足目標之量。

The *Ying Buzu* (“Excess and Deficit”) chapter presents the method of double false position, a technique for solving linear problems by making two guesses and using the errors to find the correct solution.

The term *ying* (盈) means “excess” and *buzu* (不足) means “deficit.” These represent the amounts by which a trial guess overshoots or falls short of the target.

3.1 The Method of Excess and Deficit 盈不足法

盈不足法曰：置所出率，盈不足各居其下。令維乘所出率，并以為實。并盈不足為法。實如法而一。有分者通之。盈不足相與同其買物者，置所出率，以少減多，餘以約法實。實為物價，法為人數。

詳解：

1. 設兩次出率為 a 、 b
2. 對應之盈 e_1 、不足 e_2 （或反之）
3. 人數 = $\frac{e_1 + e_2}{|a - b|}$
4. 物價 = $\frac{a \cdot e_2 + b \cdot e_1}{|a - b|}$

Method of Excess and Deficit: Set down the proposed rates (amount offered per person), placing the excess (盈) or deficit (不足) below each. Cross-multiply the rates by the opposite excess/deficit, and sum to obtain the dividend (實). Add the excess and deficit to obtain the divisor (法). Divide the dividend by the divisor. If there are fractions, convert them.

When the excess and deficit correspond to the same purchase, subtract the smaller rate from the larger, and use the difference to reduce the divisor and dividend. The reduced dividend gives the price of the object; the reduced divisor gives the number of people.

Detailed explanation:

1. Let the two proposed rates be a , b
2. Corresponding excess e_1 , deficit e_2 (or vice versa)
3. Number of people = $\frac{e_1 + e_2}{|a - b|}$
4. Price = $\frac{a \cdot e_2 + b \cdot e_1}{|a - b|}$

3.2 Problem: Buying a Chicken 共買雞

問：今有共買雞，人出九，盈十一；人出六，不足十六。問人數、雞價各幾何？

答：九人，雞價七十。

術：以盈不足法計之：

$$\text{人數} = \frac{11 + 16}{9 - 6} = \frac{27}{3} = 9$$

$$\text{雞價} = \frac{9 \times 16 + 6 \times 11}{9 - 6} = \frac{144 + 66}{3} = 70$$

驗算：每人出九錢， $9 \times 9 = 81$ ，盈 $81 - 70 = 11$ ；每人出六錢， $9 \times 6 = 54$ ，不足 $70 - 54 = 16$ 。合題。

楊輝曰：凡盈不足問題，以兩次之盈朒（盈與不足），推算人數物價，其妙在於維乘。

Problem: A group buys a chicken together. If each pays 9 *qian* (錢), there is an excess of 11 *qian*; if each pays 6 *qian*, there is a deficit of 16 *qian*. How many people are there, and what is the price of the chicken?

Answer: 9 people, price = 70 *qian*.

Method: Using the double false position formula:

$$\text{Number of people} = \frac{11 + 16}{9 - 6} = \frac{27}{3} = 9$$

$$\text{Price} = \frac{9 \times 16 + 6 \times 11}{9 - 6} = \frac{144 + 66}{3} = 70$$

Verification: At 9 each: $9 \times 9 = 81$, excess $81 - 70 = 11$. At 6 each: $9 \times 6 = 54$, deficit $70 - 54 = 16$. Checks out.

Yang Hui comments: In all excess-deficit problems, the cross-multiplication is the key technique for deriving the number of people and the price.

3.3 Problem: Buying Gold 共買金

問：共買金，人出四百，盈一貫四百；人出二百，盈四百。問人數、金價各幾何？

答：一十一人，金價九貫。

術：此為「兩盈」問題。以兩盈相減為實，兩出率相減為法：

$$\text{人數} = \frac{1400 - 400}{400 - 200} = \frac{1000}{200} = 5$$

原書答十一人，當依另一種算法。此處楊輝詳列兩種解法以資比較。

Problem: A group buys gold together. If each pays 400 *qian*, there is an excess of 1,400 *qian*; if each pays 200 *qian*, there is an excess of 400 *qian*. Find the number of people and the price of the gold.

Answer: 11 people, price = 9 *guan*.

Method: This is a “double excess” problem. Subtract the smaller excess from the larger to obtain the dividend; subtract the smaller rate from the larger to obtain the divisor:

$$\text{Number of people} = \frac{1400 - 400}{400 - 200} = \frac{1000}{200} = 5$$

The text gives 11 people, which follows a variant computation. Yang Hui presents both methods for comparison.

3.4 Problem: The Good and Inferior Fields 善田與惡田

問：善田一畝直三百，惡田七畝直五百。今買一百畝，共價一萬，問各幾何？

答：善田一十二畝半，惡田八十七畝半。

術：設善田 x 畝，惡田 y 畝。列方程組：

$$\begin{aligned} x + y &= 100 \\ 300x + \frac{500}{7}y &= 10000 \end{aligned}$$

解之：善田 = $12\frac{1}{2}$ 畝，惡田 = $87\frac{1}{2}$ 畝。

Problem: Good land costs 300 *qian* per *mu* (畝); inferior land costs 500 *qian* for 7 *mu*. If 100 *mu* are purchased for a total of 10,000 *qian*, how many *mu* of each type were bought?

Answer: Good land = $12\frac{1}{2}$ *mu*, inferior land = $87\frac{1}{2}$ *mu*.

Method: Let $x = \text{mu}$ of good land, $y = \text{mu}$ of inferior land:

$$\begin{aligned} x + y &= 100 \\ 300x + \frac{500}{7}y &= 10000 \end{aligned}$$

Solving: Good land = $12\frac{1}{2}$ *mu*, inferior land = $87\frac{1}{2}$ *mu*.

3.5 Problem: Large and Small Vessels 大器與小器

問：今有大器五、小器一，共容三斛；大器一、小器五，共容二斛。問大小器各容幾何？

術：設大器容量為 L ，小器容量為 S ：

$$\begin{aligned} 5L + S &= 3 \\ L + 5S &= 2 \end{aligned}$$

相加： $6L + 6S = 5$ ，得 $L + S = \frac{5}{6}$ 。相減： $4L - 4S = 1$ ，得 $L - S = \frac{1}{4}$ 。聯立解之： $L = \frac{13}{24}$ 斛， $S = \frac{7}{24}$ 斛。

楊輝注：此題亦可用盈不足法求解，設大器、小器各為一斛，分別求盈不足，再依術推算。

Problem: Five large vessels and one small vessel together hold 3 *hu*; one large vessel and five small vessels together hold 2 *hu*. How much does each type of vessel hold?

Method: Let $L = \text{capacity of large vessel}$, $S = \text{capacity of small vessel}$:

$$\begin{aligned} 5L + S &= 3 \\ L + 5S &= 2 \end{aligned}$$

Adding: $6L + 6S = 5$, so $L + S = \frac{5}{6}$. Subtracting: $4L - 4S = 1$, so $L - S = \frac{1}{4}$. Solving simultaneously: $L = \frac{13}{24}$ *hu*, $S = \frac{7}{24}$ *hu*.

Yang Hui's commentary: This problem can also be solved by the excess-deficit method, assuming each vessel holds 1 *hu* and computing the excess/deficit for each case.

3.6 Problem: The Horses of Liang 良馬與駑馬

問：良馬初日行一百九十三里，日增十三里；駑馬初日行九十七里，日減半里。良馬逐駑馬，問幾何日及之？

術：以盈不足法，設十五日為初試：

- 良馬十五日所行：等差數列之和 $S_{15} = \frac{15}{2}(2 \times 193 + 14 \times 13) = \frac{15}{2} \times 568 = 4260$ 里
 - 駑馬十五日所行： $S'_{15} = \frac{15}{2}(2 \times 97 - 14 \times 0.5) = \frac{15}{2} \times 187 = 1402.5$ 里
- 以盈不足術迭代，終得追及之日。

Problem: A fine horse travels 193 *li* on the first day, increasing by 13 *li* each day; an inferior horse travels 97 *li* on the first day, decreasing by $\frac{1}{2}$ *li* each day. The fine horse chases the inferior horse. After how many days will it catch up?

Method: Using the excess-deficit method, first try 15 days:

- Fine horse in 15 days: arithmetic series sum $S_{15} = \frac{15}{2}(2 \times 193 + 14 \times 13) = \frac{15}{2} \times 568 = 4260$ li
- Inferior horse in 15 days: $S'_{15} = \frac{15}{2}(2 \times 97 - 14 \times 0.5) = \frac{15}{2} \times 187 = 1402.5$ li

Iterate using the excess-deficit method until convergence is achieved.

4 Chapter on Rectangular Arrays 方程第八

方程章處理線性方程組，以矩形陣列（矩陣）排列係數，進行消元運算——西方後來稱之為高斯消去法。

「方」意為「矩形排列」，「程」意為「計算程序」。此方陣即今日所謂之矩陣。中國古代數學家發展了正負數之概念以處理方程組中之減法運算。

The *Fangcheng* (“Rectangular Arrays”) chapter deals with systems of linear equations, solved by arranging coefficients in a rectangular array (matrix) and performing elimination — the method known in the West as Gaussian elimination.

The term *fang* (方) means “rectangular arrangement” and *cheng* (程) means “calculation procedure.” This rectangular array is what we now call a matrix. Ancient Chinese mathematicians developed the concept of positive and negative numbers to handle the subtraction operations in solving these systems.

4.1 The Method of Rectangular Arrays 方程法

方程法曰：所求率互乘困行，以少減多，再求減損。物為法，實如法而一。依所問排列，逐物與固而鄰行相對。

詳解：

1. 依題意列出各物之係數及實（常數項）
2. 兩行互乘，以少減多，消去一未知數
3. 重複此過程，直至每行僅餘一未知數
4. 以係數為法，常數為實，除之得未知數之值

Method of Rectangular Arrays: Set up the coefficients of each type of grain in columns (arrays). Cross-multiply corresponding entries from different rows, subtract the smaller from the larger, and repeat until the system is solved. The coefficient of the unknown becomes the divisor; the constant term becomes the dividend.

Detailed explanation:

1. List coefficients of each commodity and the constant terms according to the problem
2. Cross-multiply two rows, subtract the smaller from the larger, eliminating one unknown
3. Repeat until each row contains only one unknown
4. Divide the constant by the coefficient to obtain the value of the unknown

4.2 Problem: Three Grades of Grain 上禾、中禾、下禾

問：今有上禾三秉、中禾二秉、下禾一秉，實三十九斗；上禾二秉、中禾三秉、下禾一秉，實三十四斗；上禾一秉、中禾二秉、下禾三秉，實二十六斗。問上、中、下禾一秉各實幾何？

答：上禾一秉九斗四分斗之一，中禾一秉四斗四分斗之一，下禾一秉二斗四分斗之三。

術：設 x 、 y 、 z 分別為上、中、下禾每秉之實：

$$3x + 2y + z = 39$$

$$2x + 3y + z = 34$$

$$x + 2y + 3z = 26$$

第一步：以式 (2) 減式 (1)： $x - y = 5$ 式 (4)

第二步：式 (2) 乘 3 減式 (3)：

$$(6x + 9y + 3z) - (x + 2y + 3z) = 102 - 26$$

$$5x + 7y = 76 \quad \text{式 (5)}$$

第三步：由式 (4)： $x = y + 5$ ，代入式 (5)：

$$5(y + 5) + 7y = 76 \Rightarrow 12y = 51 \Rightarrow y = 4\frac{1}{4}$$

第四步：回代： $x = 9\frac{1}{4}$ ， $z = 2\frac{3}{4}$ 。

楊輝圖解：以矩陣排列如下——

$$\begin{array}{ccc|c} 3 & 2 & 1 & 39 \\ 2 & 3 & 1 & 34 \\ 1 & 2 & 3 & 26 \end{array}$$

逐行相減消元，即今之「高斯消去法」。

Problem: Three bundles of top-grade grain, two bundles of middle-grade, and one bundle of low-grade yield 39 *dou* (斗). Two bundles of top-grade, three of middle-grade, and one of low-grade yield 34 *dou*. One bundle of top-grade, two of middle-grade, and three of low-grade yield 26 *dou*. How much grain does one bundle of each grade contain?

Answer: Top = $9\frac{1}{4}$ *dou*, Middle = $4\frac{1}{4}$ *dou*, Low = $2\frac{3}{4}$ *dou*.

Method: Let x , y , z be the yields of top, middle, and low grade grain:

$$3x + 2y + z = 39$$

$$2x + 3y + z = 34$$

$$x + 2y + 3z = 26$$

Step 1: Subtract (2) from (1): $x - y = 5$ Equation (4)

Step 2: Multiply (2) by 3 and subtract (3):

$$(6x + 9y + 3z) - (x + 2y + 3z) = 102 - 26$$

$$5x + 7y = 76 \quad \text{Equation (5)}$$

Step 3: From (4): $x = y + 5$. Substitute into (5):

$$5(y + 5) + 7y = 76 \Rightarrow 12y = 51 \Rightarrow y = 4\frac{1}{4}$$

Step 4: Back-substitute: $x = 9\frac{1}{4}$, $z = 2\frac{3}{4}$.

Yang Hui's diagram arranges the system as:

$$\begin{array}{ccc|c} 3 & 2 & 1 & 39 \\ 2 & 3 & 1 & 34 \\ 1 & 2 & 3 & 26 \end{array}$$

Row subtraction and elimination gives what we now call "Gaussian elimination."

4.3 Problem: Five Families Sharing a Well 五家共井

問：今有五家共井。甲二綆不足，如乙一綆；乙三綆不足，如丙一綆；丙四綆不足，如丁一綆；丁五綆不足，如戊一綆；戊六綆不足，如甲一綆。如各得所不足一綆，皆逮。問井深、綆長各幾何？

術：設井深為 d ，甲、乙、丙、丁、戊之綆長分別為 a 、 b 、 c 、 d' 、 e ：

$$\begin{aligned} 2a + b &= d & 3b + c &= d & 4c + d' &= d \\ 5d' + e &= d & 6e + a &= d \end{aligned}$$

此為五方程六未知數之不定方程組，以方程術求解。各綆長互為質數之比例。

楊輝注：此題為《九章》中最著名之問題之一，展示中國古代數學家解不定方程組之能力。

Problem: Five families share a well. Family A's 2 ropes fall short by the length of family B's 1 rope; family B's 3 ropes fall short by family C's 1 rope; family C's 4 ropes fall short by family D's 1 rope; family D's 5 ropes fall short by family E's 1 rope; family E's 6 ropes fall short by family A's 1 rope. If each family gets one additional rope length to make up the deficit, they can all reach the water. Find the depth of the well and the length of each family's rope.

Method: Let d = depth of well, a, b, c, d', e = rope lengths:

$$\begin{aligned} 2a + b &= d & 3b + c &= d & 4c + d' &= d \\ 5d' + e &= d & 6e + a &= d \end{aligned}$$

This is a system of 5 equations in 6 unknowns, solved by the method of rectangular arrays. The rope lengths are in ratios of mutually prime integers.

Yang Hui's commentary: This is one of the most famous problems in the Nine Chapters, demonstrating the Chinese mathematicians' ability to solve underdetermined systems.

4.4 Problem: Gold and Silver Weights 金與銀

問：今有金九、銀十一，共重適等。交易其一，則金輕十三兩。問金、銀一塊各重幾何？

答：金重二斤三兩一十八銖，銀重一斤一十三兩六銖。

術：設金每塊重 g 兩，銀每塊重 s 兩：

$$\begin{aligned} 9g &= 11s \\ 8g + s &= 10s + g - 13 \quad (\text{交換一塊後}) \end{aligned}$$

由第二式： $7g - 9s = -13$ 。聯立 $9g = 11s$ ：

$$g = \frac{143}{4} \text{兩} = 35.75 \text{兩}, \quad s = \frac{117}{4} \text{兩} = 29.25 \text{兩}$$

換算為斤兩銖制即可得答。

Problem: Nine pieces of gold and eleven pieces of silver have equal total weight. If one piece is exchanged between them, the gold side becomes 13 *liang* (兩) lighter. How much does each piece of gold and silver weigh?

Answer: Gold = 2 *jin* 3 *liang* 18 *zhu* (銖), Silver = 1 *jin* 13 *liang* 6 *zhu*.

Method: Let g = weight of one gold piece, s = weight of one silver piece:

$$\begin{aligned} 9g &= 11s \\ 8g + s &= 10s + g - 13 \quad (\text{after exchanging one piece}) \end{aligned}$$

From the second equation: $7g - 9s = -13$. Combined with $9g = 11s$:

$$g = \frac{143}{4} \text{liang} = 35.75 \text{liang}, \quad s = \frac{117}{4} \text{liang} = 29.25 \text{liang}$$

Convert to the *jin-liang-zhu* system to obtain the answer.

4.5 Problem: Buying Oxen, Sheep, and Pigs 牛、羊、豕

問：有賣牛二、買羊五，以買豕二，有餘錢一千；賣牛三、買豕三，以買羊三，錢適足；賣羊五、買豕五，以買牛二，錢不足六百。問牛、羊、豕價各幾何？

術：設牛價 x 、羊價 y 、豕價 z ：

$$-2x + 5y - 2z = 1000$$

$$-3x + 3y - 3z = 0$$

$$-2x + 5y + 5z = -600$$

此方程組以方程術（正負術）求解，為中國數學中最早使用正負數之例之一。

楊輝詳解：「以正負術入之，異名相除，同名相加，正無入負之，負無入正之。」

Problem: Selling 2 oxen and buying 5 sheep, then buying 2 pigs, leaves 1,000 *qian* surplus. Selling 3 oxen and 3 pigs to buy 3 sheep breaks even. Selling 5 sheep and 5 pigs to buy 2 oxen falls short by 600 *qian*. Find the price of each animal.

Method: Let x = price of ox, y = price of sheep, z = price of pig:

$$-2x + 5y - 2z = 1000$$

$$-3x + 3y - 3z = 0$$

$$-2x + 5y + 5z = -600$$

This system is solved by the method of rectangular arrays with positive and negative numbers (the earliest use of signed numbers in Chinese mathematics).

Yang Hui's detailed explanation: "Apply the positive-negative method: subtract opposite signs, add same signs; when positive has no partner, make it negative; when negative has no partner, make it positive."

5 Chapter on Right Triangles 勾股第九

勾股章專論直角三角形之性質及其在測量與工程中之應用。中國之「勾股定理」等同於西方之畢達哥拉斯定理。

「勾」為直角三角形之短直角邊（底），「股」為長直角邊（高），「弦」為斜邊（對直角之邊）。勾股術為中國古代數學中最重要之成就之一。

The *Gougu* (“Base and Height”) chapter is devoted to the properties of right triangles and their applications in surveying and measurement. The Chinese “Gougu theorem” is equivalent to the Pythagorean theorem.

The *gou* (勾) is the shorter leg (base) of the right triangle, the *gu* (股) is the longer leg (height), and the *xian* (弦) is the hypotenuse (the side opposite the right angle). The Gougu method is one of the most important achievements of ancient Chinese mathematics.

5.1 Key Formulas 勾股求弦

勾股求弦法曰：勾自乘，股自乘，并而開方除之，即弦。

$$\text{弦} = \sqrt{\text{勾}^2 + \text{股}^2}$$

弦勾求股法曰：勾自乘，減弦自乘，餘開方除之，即股。

$$\text{股} = \sqrt{\text{弦}^2 - \text{勾}^2}$$

弦股求勾法曰：股自乘，減弦自乘，餘開方除之，即勾。

$$\text{勾} = \sqrt{\text{弦}^2 - \text{股}^2}$$

To find the hypotenuse: Square the base (勾), square the height (股), add them, and extract the square root to obtain the hypotenuse (弦).

$$\text{Hypotenuse} = \sqrt{\text{base}^2 + \text{height}^2}$$

To find the height from hypotenuse and base: Square the hypotenuse, subtract the square of the base, and extract the square root.

$$\text{Height} = \sqrt{\text{hypotenuse}^2 - \text{base}^2}$$

To find the base from hypotenuse and height: Square the hypotenuse, subtract the square of the height, and extract the square root.

$$\text{Base} = \sqrt{\text{hypotenuse}^2 - \text{height}^2}$$

5.2 Problem: Finding the Hypotenuse 勾三股四弦五

問：勾三尺，股四尺，問弦幾何？

答：五尺。

術：勾自乘得九，股自乘得一十六，并得二十五，開方得五尺。

$$c = \sqrt{3^2 + 4^2} = \sqrt{9 + 16} = \sqrt{25} = 5 \text{ 尺}$$

此即著名之「勾三股四弦五」直角三角形，為所有直角三角形中最基本之例。

Problem: The base is 3 *chi* (尺) and the height is 4 *chi*. What is the hypotenuse?

Answer: 5 *chi*.

Method: Square the base to get 9, square the height to get 16; adding gives 25; extracting the square root gives 5 *chi*.

$$c = \sqrt{3^2 + 4^2} = \sqrt{9 + 16} = \sqrt{25} = 5 \text{ chi}$$

This is the famous 3-4-5 right triangle, the most fundamental example of all right triangles.

5.3 Problem: Finding the Height 弦十七步

問：弦十七步，勾八步，問股幾何？

答：十五步。

術：弦自乘得二百八十九，勾自乘得六十四，相減餘二百二十五，開方得一十五步。

$$\text{股} = \sqrt{17^2 - 8^2} = \sqrt{289 - 64} = \sqrt{225} = 15 \text{步}$$

Problem: The hypotenuse is 17 paces and the base is 8 paces. What is the height?

Answer: 15 paces.

Method: Square the hypotenuse to get 289, square the base to get 64; the difference is 225; extracting the square root gives 15 paces.

$$\text{Height} = \sqrt{17^2 - 8^2} = \sqrt{289 - 64} = \sqrt{225} = 15 \text{ paces}$$

5.4 Problem: The Reed in the Pond 葭生池中

問：池方一丈，葭生其中央，出水一尺。引葭至岸，適與岸齊。問水深、葭長各幾何？

術：設水深為 d ，葭長為 L 。葭出水面一尺，故 $L = d + 1$ 。引至岸時，葭為直角三角形之弦，勾為水深 d ，股為池半寬五尺：

$$L^2 = d^2 + 5^2$$

代入 $L = d + 1$ ：

$$(d+1)^2 = d^2 + 25 \Rightarrow 2d+1 = 25 \Rightarrow d = 12 \text{尺}, \quad L = 13 \text{尺}$$

楊輝圖解：以勾股圖繪之，葭長為弦，水深為股，半池寬為勾。

Problem: A pond is 1 *zhang* (10 *chi*) square. A reed grows at its center and extends 1 *chi* above the water. When pulled to the shore, the reed just reaches the edge. What are the depth of the water and the length of the reed?

Method: Let d = water depth, L = reed length. The reed extends 1 *chi* above water, so $L = d + 1$. When pulled to shore, the reed forms the hypotenuse of a right triangle with legs d (water depth) and 5 *chi* (half the pond width):

$$L^2 = d^2 + 5^2$$

Substituting $L = d + 1$:

$$(d+1)^2 = d^2 + 25 \Rightarrow 2d+1 = 25 \Rightarrow d = 12 \text{ chi}, \quad L = 13 \text{ chi}$$

Yang Hui's diagram: In the Gougu figure, the reed length is the hypotenuse, the water depth is the height, and half the pond width is the base.

5.5 Problem: The Door and the Pole 戶高與廣

問：戶高多於廣六尺八寸，兩隅相去適一丈。問戶高、廣各幾何？

答：高九尺六寸，廣二尺八寸。

術：設高為 h ，廣為 w ：

$$h - w = 6.8$$

$$h^2 + w^2 = 100$$

由第一式： $h = w + 6.8$ ，代入第二式：

$$(w + 6.8)^2 + w^2 = 100 \Rightarrow 2w^2 + 13.6w + 46.24 = 100$$

$$2w^2 + 13.6w - 53.76 = 0 \Rightarrow w = 2.8 \text{尺}, \quad h = 9.6 \text{尺}$$

Problem: A door's height exceeds its width by 6 *chi* 8 *cun*. The diagonal (distance between opposite corners) is exactly 1 *zhang* (10 *chi*). Find the height and width of the door.

Answer: Height = 9 *chi* 6 *cun*, Width = 2 *chi* 8 *cun*.

Method: Let h = height, w = width:

$$h - w = 6.8$$

$$h^2 + w^2 = 100$$

From the first equation: $h = w + 6.8$. Substituting:

$$(w + 6.8)^2 + w^2 = 100 \Rightarrow 2w^2 + 13.6w + 46.24 = 100$$

$$2w^2 + 13.6w - 53.76 = 0 \Rightarrow w = 2.8 \text{ chi}, \quad h = 9.6 \text{ chi}$$

5.6 Problem: The Square Inscribed in a Right Triangle 勾股容方

問：勾六步，股十一步，問容方幾何？

答：方四步（近似值為 $\frac{66}{17} \approx 3.88$ 步）。

術：勾股容方之公式：對直角之正方形邊長 s 為

$$s = \frac{ab}{a+b} = \frac{6 \times 11}{6+11} = \frac{66}{17} \approx 3.88 \text{ 步}$$

楊輝注：此術可以面積法證之。以勾、股為邊作兩正
方形，再作對角線，其交點即為容方之一角。

Problem: In a right triangle with base 6 paces and height 11 paces, find the side length of the largest square that can be inscribed with one side on the hypotenuse.

Answer: Approximately 4 paces (exact value $\frac{66}{17} \approx 3.88$ paces).

Method: For a right triangle with base a and height b , the side s of the inscribed square (with one side on the hypotenuse) is:

$$s = \frac{ab}{a+b} = \frac{6 \times 11}{6+11} = \frac{66}{17} \approx 3.88 \text{ paces}$$

Yang Hui's commentary: This method can be proved using the area method. Construct squares on the base and height, draw their diagonals; the intersection gives a corner of the inscribed square.

5.7 Problem: Measuring the Square City 邑方測量

問：邑方不知大小，各中開門。北門外二十步有木。出南門十四步，折而西行一千七百七十五步見木。問邑方幾何？

答：二百五十步。

術：設邑方為 s 。由相似三角形：

$$\frac{s/2}{20} = \frac{1775}{s+34}$$

交叉相乘： $s(s+34) = 2 \times 20 \times 1775 = 71000$

$$s^2 + 34s - 71000 = 0 \Rightarrow s = 250 \text{ 步}$$

楊輝圖解：繪邑方圖，標五門及木之位置，以相似三角形之比例求解。此題為中國古代測量學之傑作。

Problem: A square city has gates at the midpoint of each side. A tree stands 20 paces outside the north gate. Walking 14 paces out from the south gate, then turning west and walking 1,775 paces, the tree becomes visible. Find the side length of the city.

Answer: 250 paces.

Method: Let s = side of the square city. By similar triangles:

$$\frac{s/2}{20} = \frac{1775}{s+34}$$

Cross-multiplying: $s(s+34) = 2 \times 20 \times 1775 = 71000$

$$s^2 + 34s - 71000 = 0 \Rightarrow s = 250 \text{ paces}$$

Yang Hui's diagram: Draw the square city, mark the five gates and the tree's position, and solve using proportions of similar triangles. This problem is a masterpiece of ancient Chinese surveying.

6 Chapter on Construction 商功

商功（「工程估算」）章論各種幾何立體之體積計算，用於營建與土木工程。

「商」意為「估算」或「計量」，「功」意為「工程」或「工作量」。此章提供城牆、堤壩、溝渠及各種立體之體積公式。

The *Shanggong* (“Construction Consultations”) chapter deals with the calculation of volumes for various geometric solids used in engineering and construction projects.

The term *shang* (商) means “to estimate” or “to calculate,” and *gong* (功) means “work” or “construction project.” This chapter provides volume formulas for city walls, dikes, ditches, and various solids.

6.1 Volume Formulas 求積公式

商功求積法曰：穿地四尺為壤五尺，為堅三尺。此穿地求壤四之，求堅三之，皆一而一。

城垣堤溝塹渠，皆同術。并上下廣而半之，以高或深乘之，又以袤乘之，即積尺。

$$V = \frac{(w_{\text{上}} + w_{\text{下}})}{2} \times h \times L$$

其中 $w_{\text{上}}$ 為上廣， $w_{\text{下}}$ 為下廣， h 為高（或深）， L 為袤（長度）。

Volume conversion ratios: When excavating earth: 4 *chi* of loose earth becomes 5 *chi* of piled earth, or 3 *chi* of compacted earth. To convert: multiply loose volume by $\frac{5}{4}$ for piled earth, or by $\frac{3}{4}$ for compacted earth.

Method for city walls, walls, dikes, ditches, and canals: Add the top and bottom widths, halve the sum, multiply by the height (or depth), and multiply by the length to obtain the volume in cubic *chi*.

$$V = \frac{(w_{\text{top}} + w_{\text{bottom}})}{2} \times h \times L$$

where w_{top} is the top width, w_{bottom} is the bottom width, h is the height (or depth), and L is the length.

6.2 Problem: Volume of a City Wall 城垣積

問：城下廣四丈，上廣二丈，高五丈，袤一百二十六丈五尺。問積幾何？

答：一百八十九萬七千五百尺。

術：

$$V = \frac{(40 + 20)}{2} \times 50 \times 1265 = 30 \times 50 \times 1265 = 1,897,500 \text{ 立方尺}$$

(注：一丈 = 十尺。下廣四丈 = 四十尺，上廣二丈 = 二十尺，高五丈 = 五十尺。)

Problem: A city wall has bottom width 4 *zhang* (丈), top width 2 *zhang*, height 5 *zhang*, and length 126½ *zhang*. What is its volume?

Answer: 1,897,500 cubic *chi*.

Method:

$$V = \frac{(40 + 20)}{2} \times 50 \times 1265 = 30 \times 50 \times 1265 = 1,897,500 \text{ cubic } chi$$

(Note: 1 *zhang* = 10 *chi*. Bottom width 4 *zhang* = 40 *chi*, top width 2 *zhang* = 20 *chi*, height 5 *zhang* = 50 *chi*.)

6.3 Problem: Square Tower 方亭臺

問：方亭臺，上方四丈，下方五丈，高五丈。問積幾何？

答：一十萬一千六百六十六尺太半尺（即 $101,666\frac{2}{3}$ 立方尺）。

術：方亭（正截頭方錐）體積公式：

$$V = \frac{h}{3}(a^2 + ab + b^2)$$

其中 $a = 40$ 尺（上方邊長）， $b = 50$ 尺（下方邊長）， $h = 50$ 尺（高）：

$$V = \frac{50}{3}(40^2 + 40 \times 50 + 50^2) = \frac{50}{3}(1600 + 2000 + 2500) = \frac{50 \times 6100}{3} = 101,666\frac{2}{3}$$

Problem: A square platform has top side 4 *zhang*, bottom side 5 *zhang*, and height 5 *zhang*. What is its volume?

Answer: $101,666\frac{2}{3}$ cubic *chi*.

Method: For a square frustum (方亭):

$$V = \frac{h}{3}(a^2 + ab + b^2)$$

where $a = 40$ *chi* (top side), $b = 50$ *chi* (bottom side), $h = 50$ *chi* (height):

6.4 Problem: Round Tower 圓亭臺

問：圓亭臺，上周二丈，下周三丈，高二丈。問積幾何？

術：圓亭（正截頭圓錐）體積公式：

$$V = \frac{h}{36}(C_1^2 + C_1 \cdot C_2 + C_2^2) \times \frac{4}{\pi^2}$$

其中 C_1 、 C_2 分別為上下周長。以中國古率 $\pi \approx 3$ 計之：

$$V \approx \frac{h}{12}(r_1^2 + r_1 r_2 + r_2^2)$$

此處上周 $C_1 = 20$ 尺，下周 $C_2 = 30$ 尺，高 $h = 20$ 尺。

楊輝注：「圓亭之術，以周徑相乘，以高乘之，三十六而一。」

Problem: A circular platform has top circumference 2 *zhang*, bottom circumference 3 *zhang*, and height 2 *zhang*. What is its volume?

Method: For a circular frustum (圓亭):

$$V = \frac{h}{36}(C_1^2 + C_1 \cdot C_2 + C_2^2) \times \frac{4}{\pi^2}$$

where C_1 and C_2 are the top and bottom circumferences. In ancient Chinese mathematics, $\pi \approx 3$:

$$V \approx \frac{h}{12}(r_1^2 + r_1 r_2 + r_2^2)$$

Here $C_1 = 20$ *chi*, $C_2 = 30$ *chi*, $h = 20$ *chi*.

Yang Hui's commentary: "For a circular platform, multiply the circumference and diameter, multiply by the height, and divide by 36."

6.5 Problem: Grain Pile on the Ground 委粟平地

問：委粟平地，下周一十二丈，高二丈。問積幾何？及為粟各幾何？

術：粟堆為圓錐形。以中國古率 $\pi \approx 3$ ：

$$V = \frac{C^2 \times h}{36} = \frac{120^2 \times 20}{36} = \frac{288000}{36} = 8000 \text{ 立方尺}$$

再以粟之容積率換算為斛數。

楊輝注：委粟於平地成圓錐，其下周即圓錐之底周，高即圓錐之高。以周求積，三十六而一，即古率之術也。

Problem: A pile of millet grain on flat ground has base circumference 12 *zhang* and height 2 *zhang*. What is its volume and how many *hu* of millet does it contain?

Method: The pile forms a cone. Using the formula with $\pi \approx 3$:

$$V = \frac{C^2 \times h}{36} = \frac{120^2 \times 20}{36} = \frac{288000}{36} = 8000 \text{ cubic } chi$$

The conversion to *hu* of grain uses the standard conversion rate.

Yang Hui's commentary: Millet piled on flat ground forms a cone; its base circumference is the cone's base circumference, and its height is the cone's height. Using the circumference to find the volume and dividing by 36 is the method of the ancient rate.

7 Yang Hui's Triangle 開方作法本源

楊輝最著名之貢獻之一，為保存現今稱為「楊輝三角」（西方常稱帕斯卡三角）之數表。楊輝將此歸功於北宋數學家賈憲（約 1010–1070 年）。

此三角列出二項式展開之係數，用於開方運算。楊輝稱之為「開方作法本源」，意為「開方運算之根本方法」。

One of Yang Hui's most celebrated contributions is his preservation of what is now known as “Yang Hui's Triangle” (often called Pascal's Triangle in the West). Yang Hui attributes this to the Northern Song mathematician Jia Xian (c. 1010–1070).

This triangle lists the coefficients for binomial expansion, used in root extraction. Yang Hui called it the *Kaifang zuofa benyuan* (“Source of the Method for Extracting Roots”), meaning “the fundamental method for root extraction.”

Yang Hui's Triangle 楊輝三角

				1			
				1		1	
			1		2		1
		1		3		3	
	1		4		6		4
1		5		10		10	
							1

開方作法本源：左禡乃積數，右禡乃隅算，中藏者皆廉。以廉乘商方，命實而除之。

詳解：

- 左斜邊（左禡）：皆為 1，即各次方之積數
- 右斜邊（右禡）：皆為 1，即隅算
- 中間各數（廉）：即二項式係數
此數表提供二項展開之係數：

$$(a + b)^1 = a + b$$

$$(a + b)^2 = a^2 + 2ab + b^2$$

$$(a + b)^3 = a^3 + 3a^2b + 3ab^2 + b^3$$

$$(a + b)^4 = a^4 + 4a^3b + 6a^2b^2 + 4ab^3 + b^4$$

$$(a + b)^5 = a^5 + 5a^4b + 10a^3b^2 + 10a^2b^3 + 5ab^4 + b^5$$

Explanation: The left side gives the coefficients for root extraction, the right side gives the corner value (always 1), and the center entries are the “shoulders” (intermediate coefficients).

Detailed explanation:

- Left diagonal (左禡): all 1s, the accumulated numbers
- Right diagonal (右禡): all 1s, the corner values
- Center entries (廉): the binomial coefficients

The triangle provides the coefficients for binomial expansion:

$$(a + b)^1 = a + b$$

$$(a + b)^2 = a^2 + 2ab + b^2$$

$$(a + b)^3 = a^3 + 3a^2b + 3ab^2 + b^3$$

$$(a + b)^4 = a^4 + 4a^3b + 6a^2b^2 + 4ab^3 + b^4$$

$$(a + b)^5 = a^5 + 5a^4b + 10a^3b^2 + 10a^2b^3 + 5ab^4 + b^5$$

7.1 Application to Root Extraction 開方之應用

楊輝以這些係數運用於增乘開方法（「遞增乘法」）以求平方根、立方根及更高次方根。此方法等同於西方數學中之霍納法（Horner's method）。

開平方示例：求 $\sqrt{55225}$

1. 置 55225 為實，從末位起每隔一位作節：5 52 25
 2. 初商：最大整數平方不大於 5 者為 2， $2^2 = 4$ ，餘 1
 3. 次商：以 20 乘初商得 40 為廉法，次商 5， $40 \times 5 = 2025$ ，餘 525
 4. 三商：以 20 乘已得數 235 得 4700，三商 5， $4705 \times 5 = 23525$ ，餘 0
- 故 $\sqrt{55225} = 235$ 。

楊輝圖解：以三角之第二行係數（1, 2, 1）對應開平方之各級運算。

Yang Hui used these coefficients in the method of *zengcheng kaifang fa* (“Increasing Multiplication Method”) for extracting square roots, cube roots, and higher-order roots. This method is equivalent to Horner's method in Western mathematics.

Square root example: Find $\sqrt{55225}$

1. Set 55225 as the dividend, mark off pairs from the right: 5 52 25
2. First digit: largest integer whose square ≤ 5 is 2, $2^2 = 4$, remainder 1
3. Second digit: multiply first result (2) by 20 to get 40; second digit is 5; $40 \times 5 = 2025$, remainder 525
4. Third digit: multiply current result (23) by 20 to get 4600; third digit is 5; $465 \times 5 = 2325$

Thus $\sqrt{55225} = 235$.

Yang Hui's diagram: The second row of the triangle (1, 2, 1) corresponds to the levels of computation in square root extraction.

Appendix: Classification of Nine Chapters Problems 九章纂類

楊輝將《九章算術》之二百四十六題按其解法分為九大類：

Yang Hui reorganized the 246 problems of the Nine Chapters into nine categories based on their solution methods:

No.	Category 類	Problems 題數
1	乘除 (Multiplication and Division)	41
2	分率 (Rates and Proportions)	41
3	合率 (Combined Rates)	10
4	互換 (Exchange and Conversion)	11
5	衰分 (Proportional Distribution)	18
6	疊積 (Stacked Accumulations)	27
7	盈不足 (Excess and Deficit)	20
8	方程 (Rectangular Arrays)	19
9	勾股 (Right Triangles)	24
Total 總計		246

楊輝竊見九章舊本，作立題法，迫圓古序云：靖康以來，罕有舊本，間有存者，狃於末習，不循本意，以至具術淹廢，偽本滋興。此說固然，殊不知所傳之本亦不得其真矣。

Yang Hui observed that in old editions of the Nine Chapters, since the Jingkang era (1126 CE), original copies have been rare. Those that survived were corrupted by later practices, deviating from the original intent, leading to the decline of proper methods and the proliferation of erroneous editions.

Yang Hui laments that since the fall of the Northern Song, authentic editions have become increasingly scarce, and even surviving copies have been corrupted by accumulated errors.

楊輝數學著作年表

- 1261 年：《詳解九章算法》十二卷
- 1262 年：《日用算法》二卷
- 1274 年：《乘除通變本末》三卷
- 1275 年：《田畝比類乘除捷法》二卷
- 1275 年：《續古摘奇算法》二卷

楊輝為中國數學史上最重要之數學家之一，其著作保存了大量已失傳之早期數學方法，為研究宋代數學之珍貴文獻。

Chronology of Yang Hui's Mathematical Works

- 1261 CE: *Xiangjie Jiuzhang Suanfa* (12 volumes)
- 1262 CE: *Riyong Suanfa* (Daily Mathematics, 2 volumes)
- 1274 CE: *Chengchu Tongbian Benmo* (3 volumes)
- 1275 CE: *Tianmu Bilei Chengchu Jiefa* (2 volumes)
- 1275 CE: *Xugu Zhaiqi Suanfa* (2 volumes)

Yang Hui is one of the most important mathematicians in Chinese history. His works preserved numerous early mathematical methods that are now lost, making them invaluable documents for the study of Song dynasty mathematics.

End of Part II 第二部分終

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